

Probability and Statistics for Biologists

DATA405 Genomic Data Science • Lecture 11 • Sep 29, 2026

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From a coin flip to DESeq2, RSEM and ChIP-seq peak calling — every concept by example

Where we are

- **Yesterday:** Python and R. We ended with a t-test on one gene (SPARCL1, $p = 0.00003$) and a warning that it is the wrong tool for 19,271 genes.
- **Today:** the ideas behind that warning — probability, the p-value, the multiple-testing burden, the false discovery rate — and how three standard tools (RSEM, DESeq2, peak callers) use them.
- **No formulas required.** Every concept is introduced with a coin, a grid, or a real count from the airway data, and then named.
- **Tomorrow:** Lab 1 — you run the coin experiment, the 19,271-tests trap and the BH correction yourself.

Today in one sentence

A sequencer does not measure a gene. It **draws reads at random** from a pool of molecules — like flipping coins and scattering rice.

So every number in genomics is a count from a random sample, and every claim ('this gene changed', 'this is a peak') must ask: could chance alone have produced this?

Coin → binomial. Rice on a grid → Poisson. Four different people → negative binomial. That is most of it.

The plan

Part	The example	The concept it teaches
1 Randomness	a coin; reads at a SNP; rice on a grid; four donors	binomial · Poisson · negative binomial
2 The p-value	is this coin fair? is SPARCL1 changed?	null hypothesis, p-value, what p is not
3 Many tests	20 candy colours; 19,271 genes with no effect	multiple-testing burden, Bonferroni, FDR, Benjamini–Hochberg
4 Best guess	54 of 138 reads; a read that fits two transcripts	likelihood, uncertainty, the EM idea
5 RNA-seq EM	your BIOL419 slides + EM in ten lines	RSEM
6 DESeq2	airway: 4 donors ± dexamethasone	normalization, NB model, borrowing strength, padj
7 ChIP-seq	your BIOL419 slides + Xu et al. 2021	Poisson peaks, mock IP vs input, posterior

Part 1

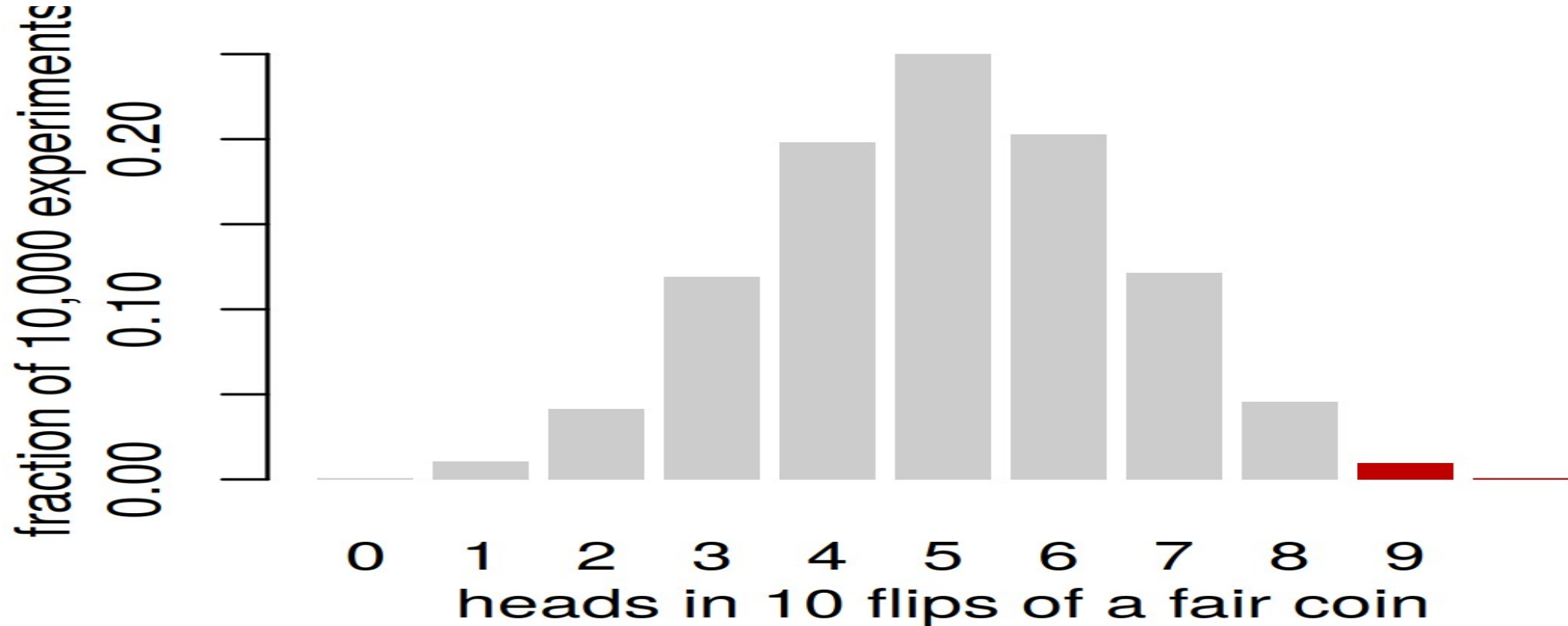
Randomness, three ways

A coin, a grid of rice, and four different people.

Start with a coin

- Flip a fair coin 10 times. You expect 5 heads — but you will often get 4, 6, sometimes 3 or 7, occasionally 9.
- Repeat the 10-flip experiment 10,000 times and count how often each result appears. That picture is the binomial distribution.
- **Probability** of an outcome = the fraction of times it happens in the long run.
- Nothing about the coin changed between experiments. The spread is what randomness looks like.

10,000 experiments of 10 flips

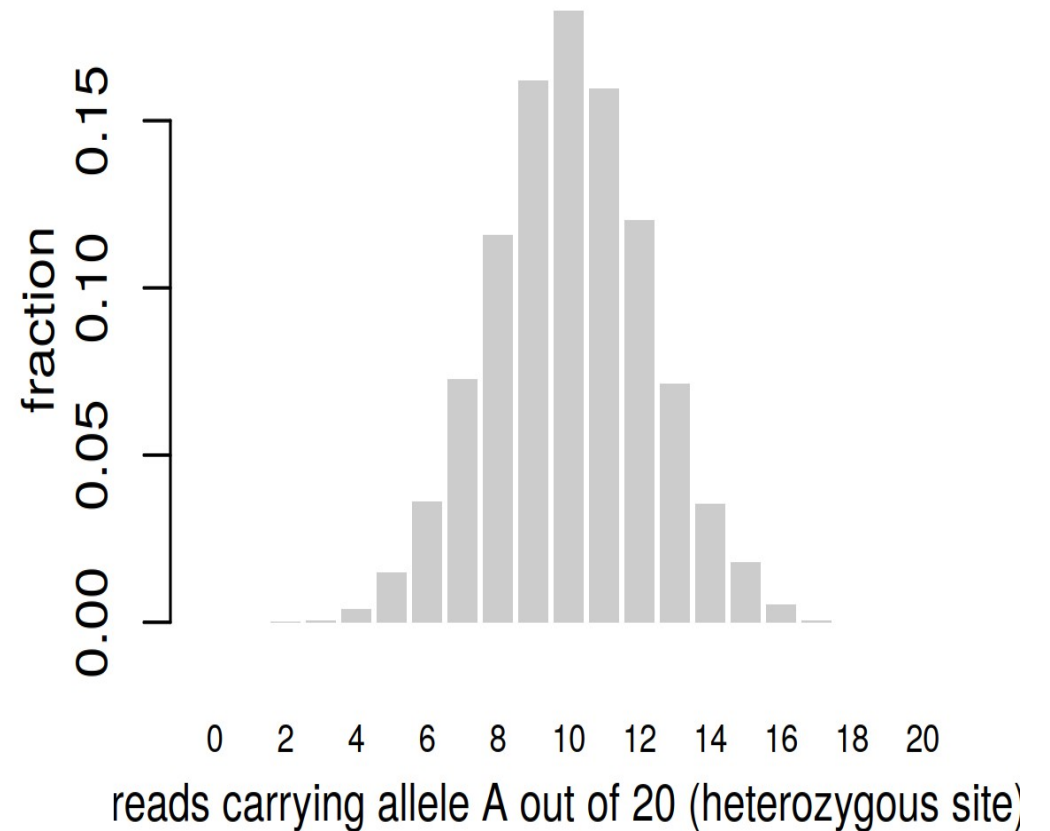


Binomial(10, 0.5). Red: 9 or 10 heads — about 1 experiment in 100.

Remember this shape. Most results sit near the expectation; extreme results are rare but they DO happen with no cause at all.

A read at a heterozygous site is a coin flip

- A person carries allele A on one chromosome and allele G on the other. Each read that covers the site comes from one chromosome or the other — 50 : 50, like a coin.
- 20 reads → you expect 10 A's, but 7 or 13 are common (see the next slide).
- **So:** if you see 12 A's out of 20, that is not evidence of anything. If you see 20 out of 20, the person is almost certainly homozygous — or the reads came from somewhere else.



Binomial(20, 0.5): allele counts at a heterozygous site

Coin intuition

YOUR TURN At a heterozygous site with 9 reads, all 9 carry the alternative allele. Roughly how often does a fair coin give 9 heads out of 9?

- A. about 1 in 10
- B. about 1 in 100
- C. about 1 in 500
- D. about 1 in 10,000

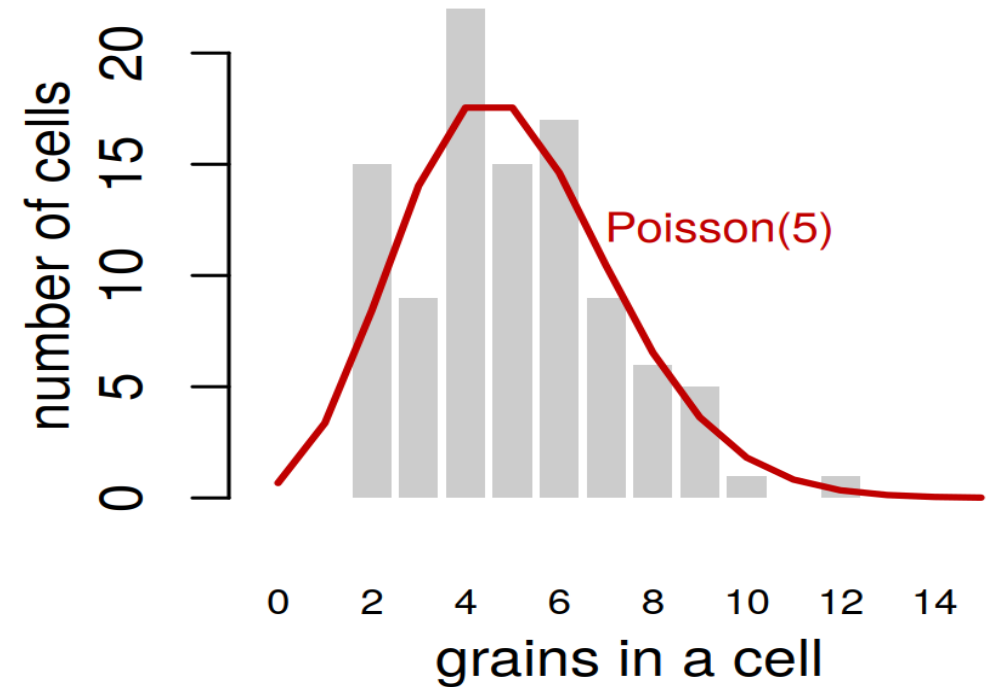
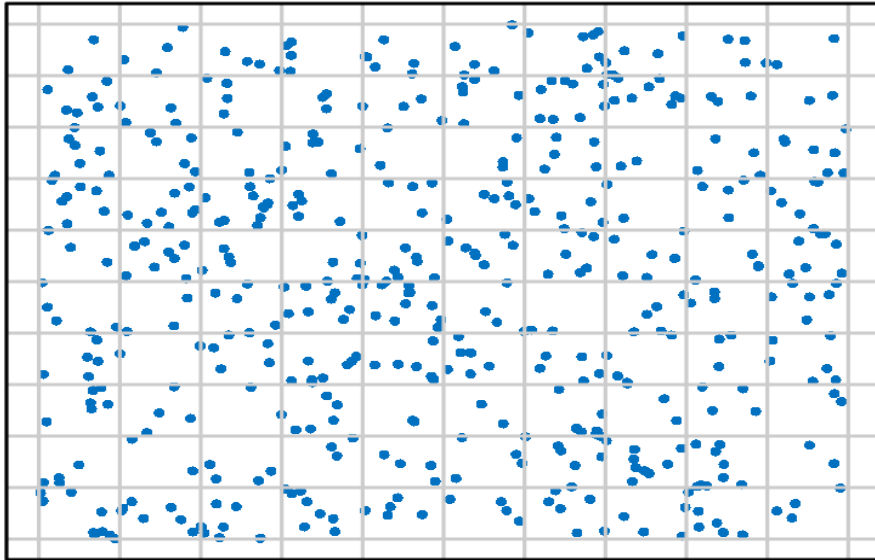
ANSWER C. $(1/2)^9 = 1/512$. Rare enough to doubt 'heterozygous' — in the lab table, such sites are mostly reads from a paralog mapping to chr22.

Scatter rice on a grid: the Poisson

- Drop 500 grains at random on a 10×10 grid. Each cell expects 5 grains — but some get 1, some get 10.
- Reads scattering across the genome behave the same way: a 500-bp window that averages 5 reads will sometimes show 10 with no biology behind it.
- **The Poisson rule of thumb:** spread $\approx \sqrt{\text{mean}}$. Mean 5 $\rightarrow$ typical wobble ± 2 . Mean 100 $\rightarrow \pm 10$. Mean 10,000 $\rightarrow \pm 100$ (1 %).
- This is why deeper sequencing gives more precise numbers, and why a 25-read window against a 5-read background is remarkable (Part 7).

500 grains, 100 cells

500 grains, 10 x 10 grid (5 per cell)



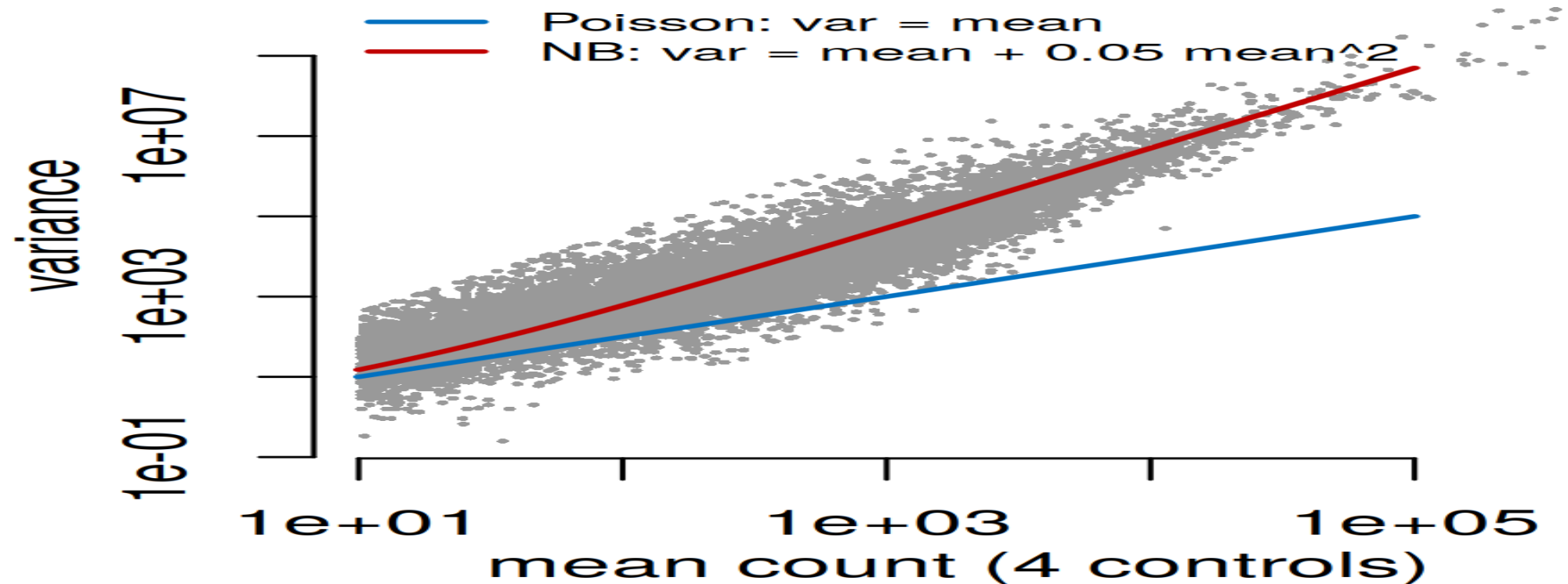
Left: the grid. Right: how many cells got 0, 1, 2, ... grains; the red curve is Poisson(5).

Uniform scattering + counting = Poisson. Any peak caller starts here.

Four donors are four different coins: the negative binomial

- Sequence the same library twice: the counts differ only by the rice-scatter wobble (Poisson).
- Sequence four different people: each has their own true expression level. The counts differ by the wobble PLUS the person-to-person difference.
- **Negative binomial = Poisson with an extra spread term:** $\text{variance} = \text{mean} + \alpha \cdot \text{mean}^2$. The number α ('dispersion') measures the biology, not the sequencer.
- Look at the four control samples of the airway experiment. Is the spread Poisson?

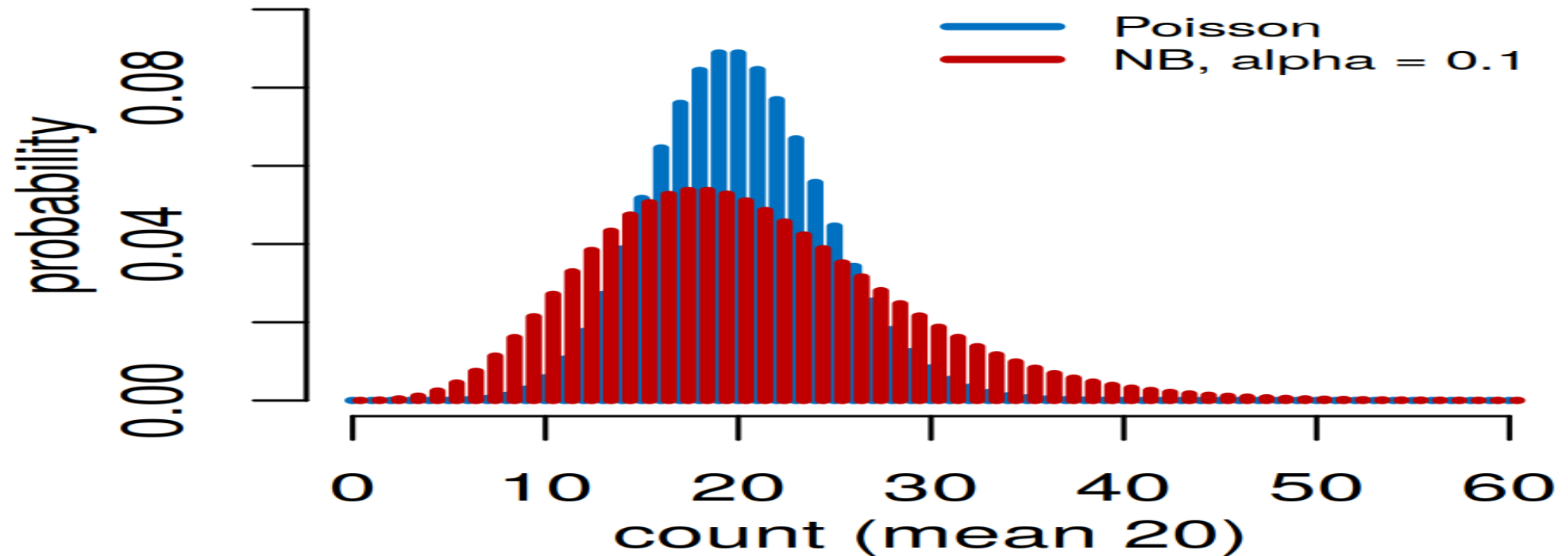
Real data: variance vs mean across 4 control samples



Each dot is a gene. Blue line: Poisson (variance = mean). Red: negative binomial, $\alpha = 0.05$.

95 % of genes lie above the Poisson line. Treat them as Poisson and every donor difference looks like a discovery. DESeq2's first job is to estimate α .

Same mean, different spread



Mean 20. Poisson: typical wobble ± 4.5 . NB with $\alpha = 0.1$: ± 6.9 . A count of 35 is a surprise under Poisson and ordinary under NB.

Which distribution you assume decides what counts as surprising. That is the whole game.

Which distribution?

YOUR TURN Match each question to coin (binomial), rice (Poisson) or donors (negative binomial).

- 1. Of 40 reads covering rs4680, 18 carry A. Heterozygous?
- 2. A 1-kb window received 60 ChIP reads; the input averages 12. Peak?
- 3. COMT has 8, 11, 6, 9 counts in four controls and 30, 25, 41, 28 in four treated. Changed?

ANSWER 1 coin (binomial, $n = 40$, $\theta = 0.5$) · 2 rice (Poisson, $\lambda = 12$) · 3 donors (negative binomial — replicates are people).

Part 2

The p-value, by example

Is this coin fair? Is this gene changed? The same question, the same answer.

Is this coin fair?

- I flip a coin 10 times and get 9 heads. Is it a trick coin?
- **Step 1 — assume the boring explanation:** the coin is fair. (The null hypothesis.)
- **Step 2 — ask how often a fair coin gives a result at least this extreme:** 9 or 10 heads (and, two-sided, 0 or 1 heads). From the picture: about 2 in 100.
- **Step 3 — that number is the p-value:** $p \approx 0.02$. Small p = 'chance alone rarely does this'.
- The p-value is a statement about the data assuming the null; it is NOT the probability that the coin is fair.

p-value = the probability of data at least this surprising IF nothing is going on.

The same computation in one line

R

```
binom.test(9, 10, 0.5)$p.value      # 9 heads in 10 flips  
# → 0.021  
  
binom.test(54, 138, 0.5)$p.value    # 54 alt reads of 138 at one site  
# → 0.013    a little low for heterozygous: allelic imbalance? mapping bias?  
  
binom.test(9, 9, 0.5)$p.value       # 9 of 9 alt reads  
# → 0.0039    not a heterozygous site
```

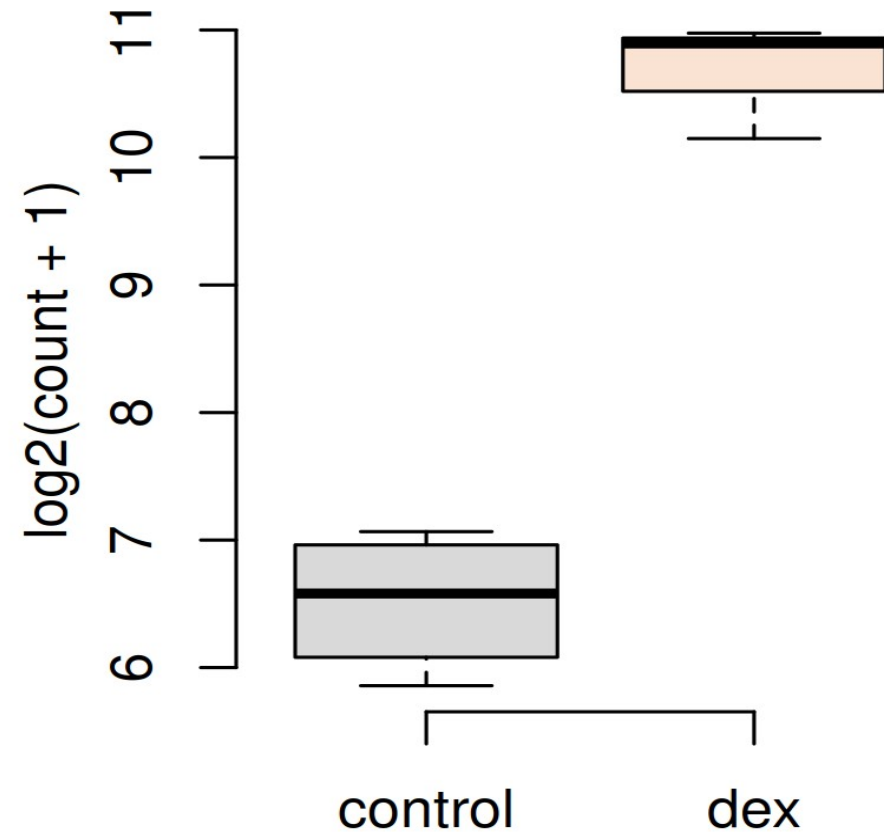
Tomorrow you run these on the real mismatch table. Same logic, real reads.

What $p = 0.05$ means — and what it does not

- **$p < 0.05$ is a convention** (Fisher, 1925): 'rarer than 1 in 20 under the null'. Nothing magical happens at 0.049 vs 0.051.
- **p is not the probability the null is true.** It is the probability of the data given the null.
- **p says nothing about size.** With enough reads, a 1 % change gets $p < 0.001$. With 4 donors, a 16-fold change can have $p = 0.2$.
- **One test, one p .** Run 20 tests and $p < 0.05$ loses its meaning — Part 3.

From a coin to a gene: the t-test is the same logic

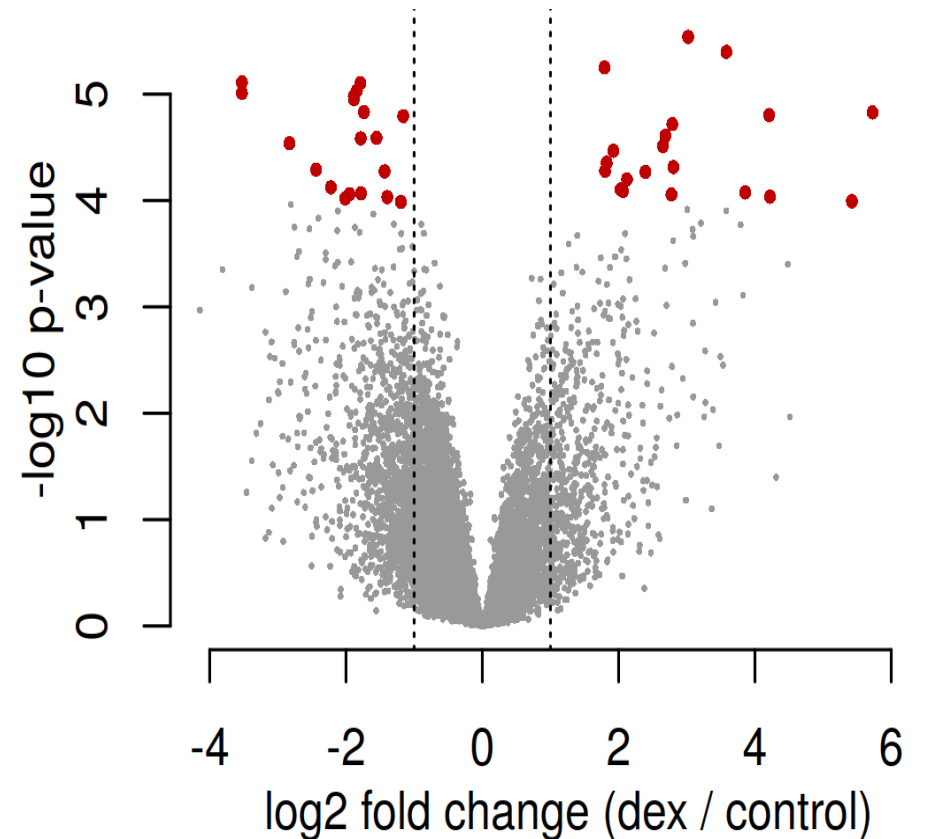
- SPARCL1: four control samples ≈ 6.5 ($\log_2$ counts), four treated ≈ 10.7 . Is the difference real?
- **Null:** the drug does nothing; the eight numbers are one pile split at random.
- **Test statistic:** difference of means $\div$ the wobble you would expect ($t = 12.5$).
- **p-value:** how often random splitting gives a t this large: 0.00003.
- Yesterday's one-line `t.test(b, a)` did exactly Steps 1–3 of the coin.



SPARCL1, $\log_2(\text{count} + 1)$, control vs dexamethasone

Effect size vs p-value: keep both

- **Effect size:** how big is the change? (log2 fold change: +4 means 16×.)
- **p-value:** how surprising is it under the null? (depends on the effect AND the noise AND the sample size)
- A volcano plot shows both: x = effect, y = $-\log_{10} p$. Genes to care about are far right or left AND high.
- A tiny effect with $p = 10^{-10}$ is real but may be biologically irrelevant. A huge effect with $p = 0.2$ in 4 donors deserves more replicates, not the wastebasket.



Airway t-tests: 19,271 genes. Red = survive the correction of Part 3.

Interpreting p

YOUR TURN A gene has $p = 0.03$. Which statement is correct?

- A. There is a 3 % chance the gene is not really changed
- B. If the gene were not changed, data this extreme would arise about 3 % of the time
- C. The gene is changed by 3 %
- D. The result will replicate 97 % of the time

ANSWER B only. A is the most common misreading (it reverses the conditional). C confuses p with effect size. D is not what p measures.

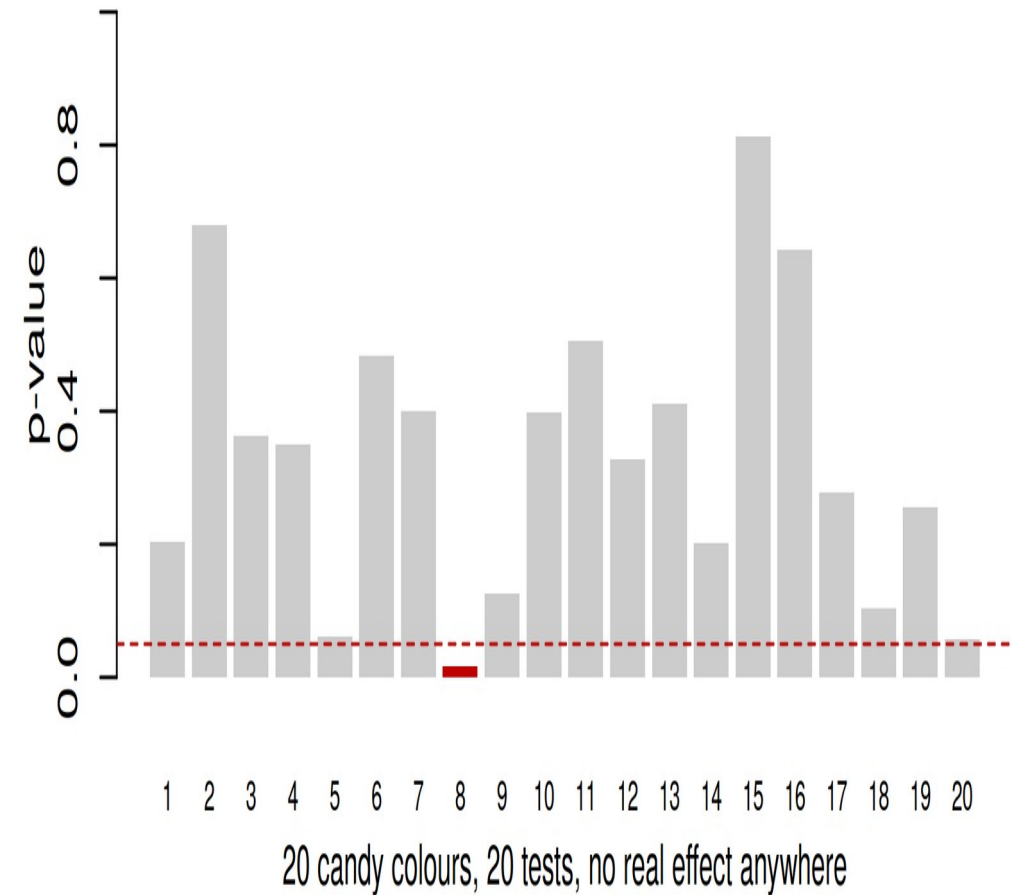
Part 3

Many tests: the burden and the false discovery rate

Twenty candy colours, 19,271 genes, and how to keep a list you can trust.

The candy experiment

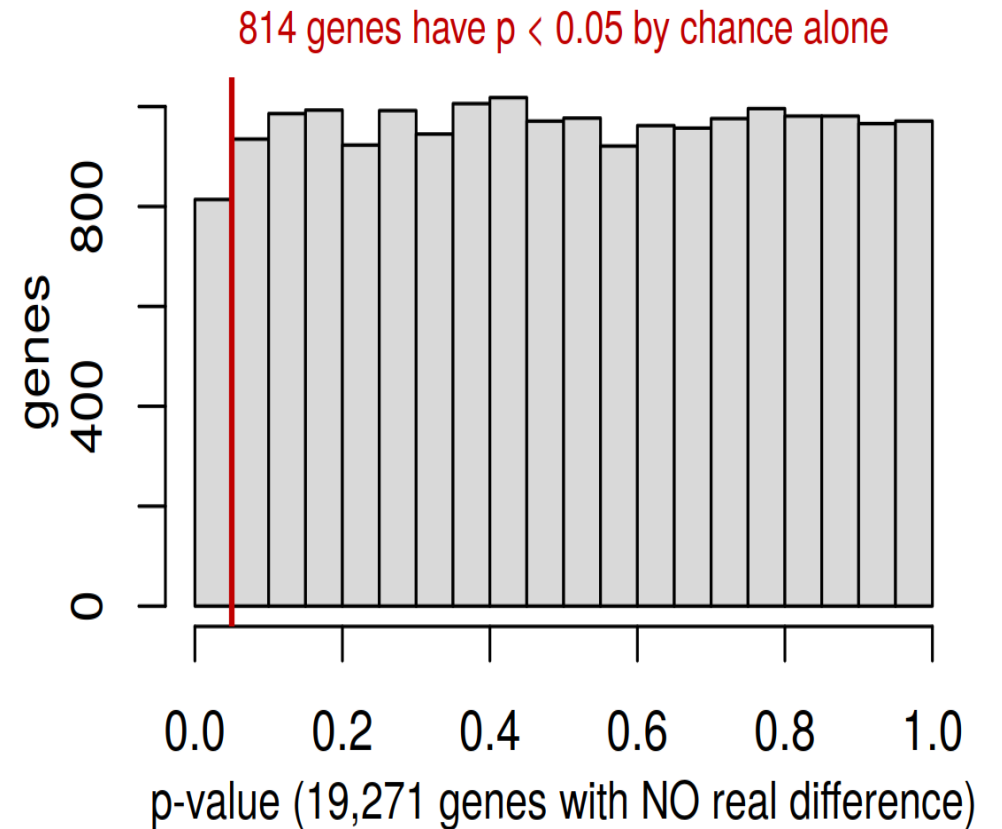
- A rumour: candy causes acne. A study tests it — no effect, $p = 0.35$.
- 'Maybe only one colour causes it?' Test 20 colours separately. One comes out $p < 0.05$. Headline: GREEN CANDY LINKED TO ACNE.
- **What happened:** 20 tests at $p < 0.05$, each with a 1-in-20 chance of a false alarm, gives a 64 % chance that at least one colour 'wins' by luck alone.
- Nobody cheated. The burden of many tests did it.



20 p-values from 20 tests with no real effect. One falls below 0.05.

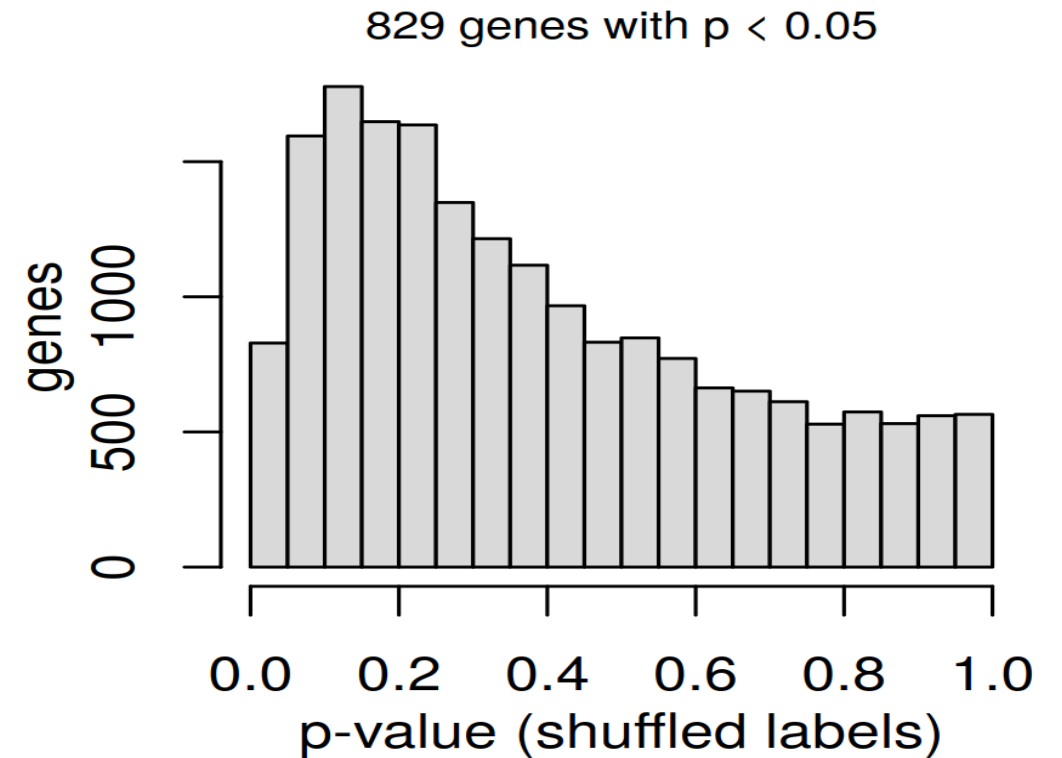
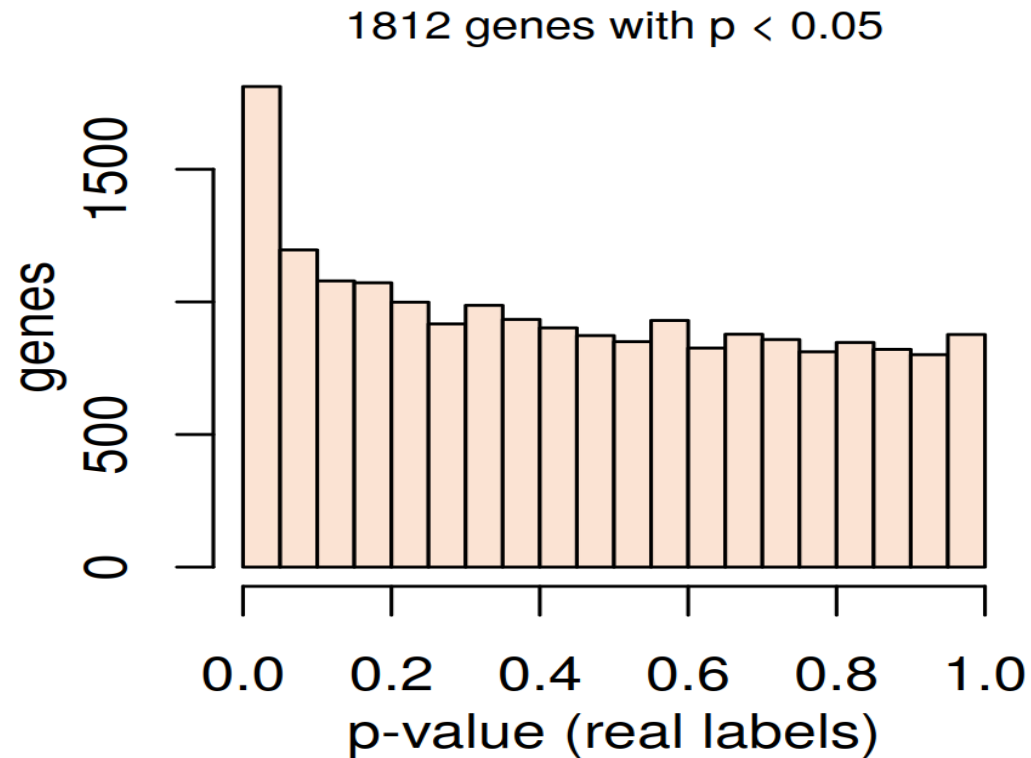
Now do it 19,271 times

- RNA-seq tests every gene. If NO gene were affected, p-values would be spread evenly between 0 and 1 (a flat histogram).
- 5 % of a flat histogram lies below 0.05 → about 800–960 genes 'significant' with nothing going on.
- Yesterday's real t-test found 1,812 genes at $p < 0.05$. Roughly half of them are the flat floor.
- **This is the multiple-testing burden:** the more questions you ask, the more wrong answers you are guaranteed.



Simulated: 19,271 genes with no effect, 4 vs 4

Real labels vs shuffled labels (airway)



Left: dex vs control. Right: the same test after shuffling which sample is which.

The spike near 0 on the left is signal; its height above the flat floor is roughly how many real discoveries there are. The right panel is the burden made visible.

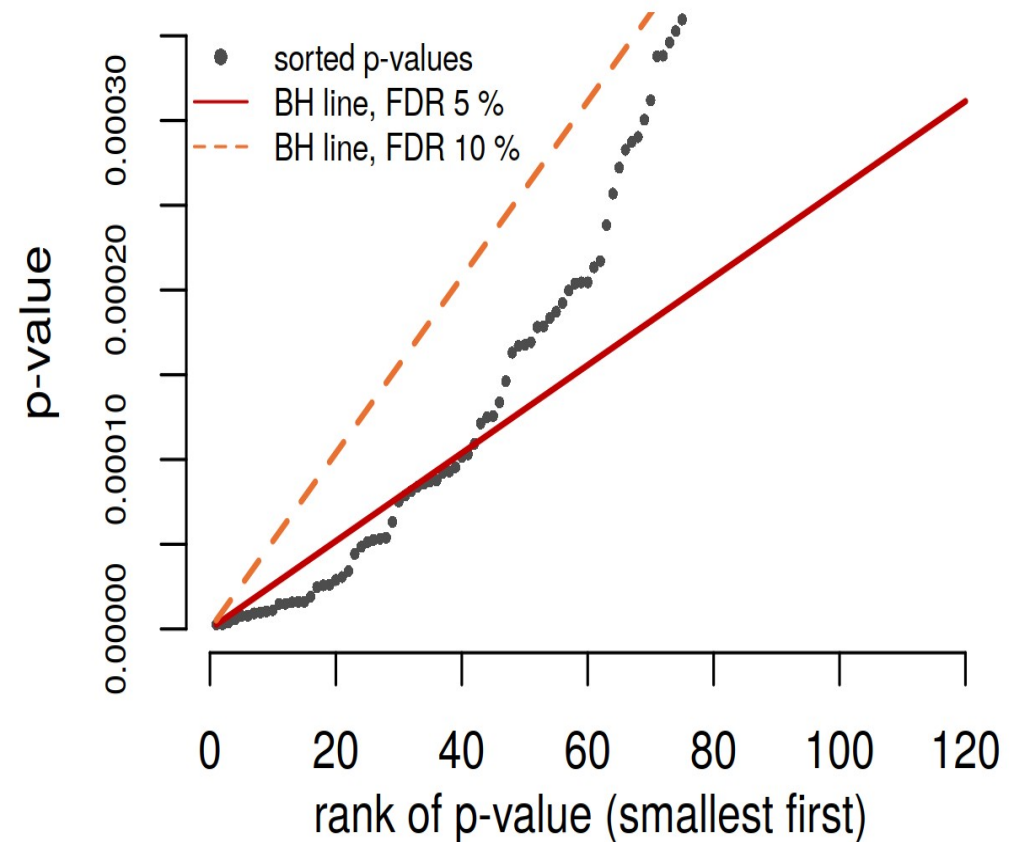
Two ways to pay the burden

- **Bonferroni — control the chance of ANY false alarm.** Divide 0.05 by the number of tests: $0.05 / 19,271 = 0.0000026$. Safe, but throws away almost everything. Right for a handful of tests (20 candy colours), wrong for a genome.
- **False discovery rate (FDR) — control the FRACTION of false alarms in your list.** 'Of the genes I call, about 5 % are expected to be false.' A statement about the list, not about one gene.
- Genomics uses FDR because a list of 2,000 genes with 100 mistakes is useful; a list of 3 genes with 0 mistakes usually is not.

Bonferroni protects one claim. FDR protects a list. Genomics makes lists.

Benjamini-Hochberg in three moves (1995)

- **1 Sort** the 19,271 p-values, smallest first:
 $p(1) \leq p(2) \leq \dots \leq p(n)$.
- **2 Draw the line** $q \cdot k / n$ for rank k : at FDR $q = 0.05$ the first p may be up to $0.05/19,271$, the 100th up to $100 \times$ that, and so on.
- **3 Keep everything up to the last p-value that is under its line.**
- Intuition: a tiny p-value is only allowed to be 'tiny' if you have few of them; the k -th smallest is judged against what k -th smallest a null experiment would produce.
- In R: `p.adjust(p, method = "BH")` returns the 'q-value' (`padj`): the smallest FDR at which that gene would be kept.



Airway t-test p-values, smallest 120, against the BH lines

BH by hand — and what it costs

R

```
p <- apply(logc, 1, function(x) tt(x[trt], x[ctrl])) # 19,271 t-tests
sum(p < 0.05, na.rm = TRUE) # → 1812 raw

q <- p.adjust(p, method = "BH") # Benjamini–Hochberg
sum(q < 0.05, na.rm = TRUE) # → 41 FDR 5 %
sum(q < 0.10, na.rm = TRUE) # → 93 FDR 10 %

0.05 / length(p) # → 2.6e-06 Bonferroni threshold
sum(p < 0.05 / length(p), na.rm = TRUE) # → 0
```

1,812 → 41. The t-test on 4 vs 4 simply has little power once the burden is paid. DESeq2 (Part 6) recovers thousands — legitimately.

Reading padj

YOUR TURN A DESeq2 table lists a gene with $p = 0.001$ and $padj = 0.08$. Your list uses FDR 0.05. In or out — and what does 0.08 mean?

- A. In: $p < 0.05$
- B. Out: at FDR 5 % this gene is not kept; if you accepted an 8 % false-discovery list, it would be
- C. Out: there is an 8 % chance it is a false positive

ANSWER B. $padj$ is the FDR level at which the gene enters the list. C is the tempting misreading: 8 % is about the list, not this gene.

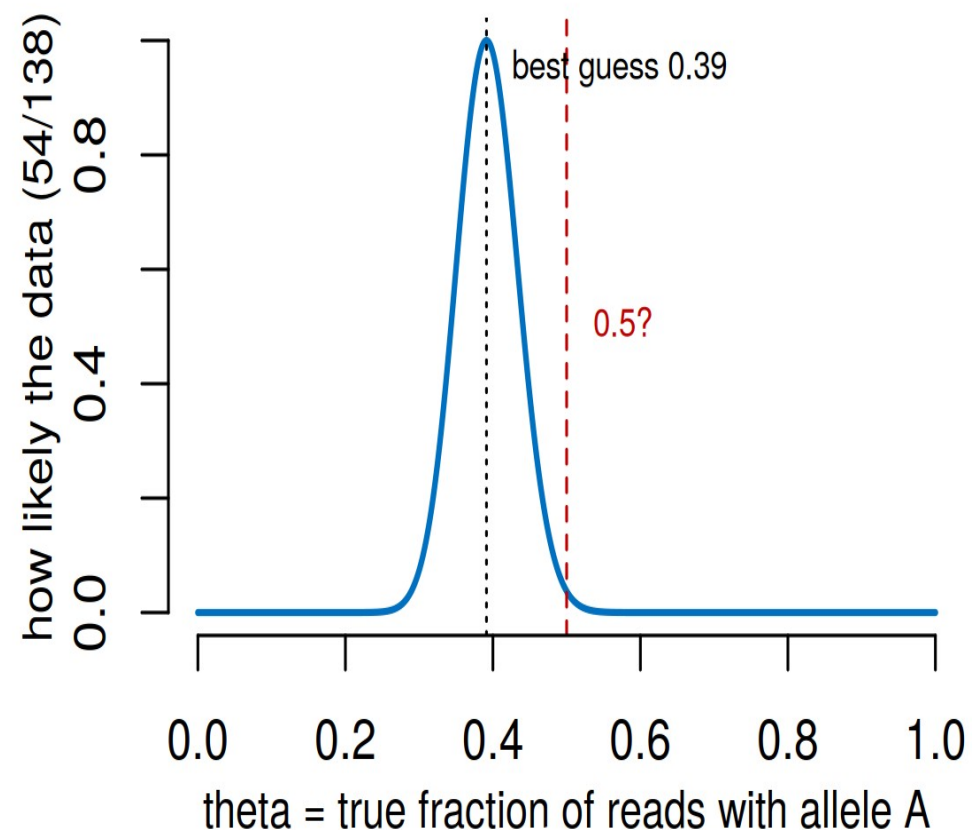
Part 4

The best guess, and hidden data

Likelihood, uncertainty, and the idea behind EM.

What is the best estimate?

- 54 of 138 reads carry allele A. What fraction θ of the molecules carry A?
- **Likelihood:** for each candidate θ , how probable are 54 of 138? Plot it; the top of the curve is the maximum-likelihood estimate: $\hat{\theta} = 54/138 = 0.39$.
- The obvious answer — but now we also see how confident to be: the curve is wide for 9 reads, narrow for 138, needle-sharp for 10,000.
- Every number in genomics — TPM, fold change, peak height — is an estimate with a curve like this behind it.



Likelihood of 54/138 for each θ . Red: the heterozygous value 0.5.

Turning the curve into a probability: Bayes

- Normalize the curve so it sums to 1 and it becomes a probability distribution for θ — with a flat prior, a Beta(55, 85).
- **Now you may ask:** $P(\theta > 0.5 \mid \text{data}) \approx 0.005$. A p-value never says this; a posterior does.
- The mock-IP paper (Part 7) asks exactly this kind of question about a peak: $P(\text{IP more enriched than input AND more than mock})$?

Likelihood: 'which θ fits best'. Bayes: 'how probable is each θ '. Same curve, different question.

Hidden data: the EM idea with two jars

- Two jars of coins: jar A's coins land heads 70 %, jar B's 30 %. Someone draws a coin from a random jar, flips it 10 times, writes down the heads, and repeats — but never tells you which jar.
- You want to estimate each jar's bias. Circular: to estimate a jar you need to know which flips came from it; to guess the jar you need its bias.
- **EM breaks the circle:** guess the biases → assign each record to the more likely jar (E-step) → re-estimate the biases from the assignments (M-step) → repeat until nothing changes.
- Each round can only make the data more likely; the answer settles in a few rounds.

Replace 'jar' by 'transcript' and 'flips' by 'a read': that is RSEM.

Part 5

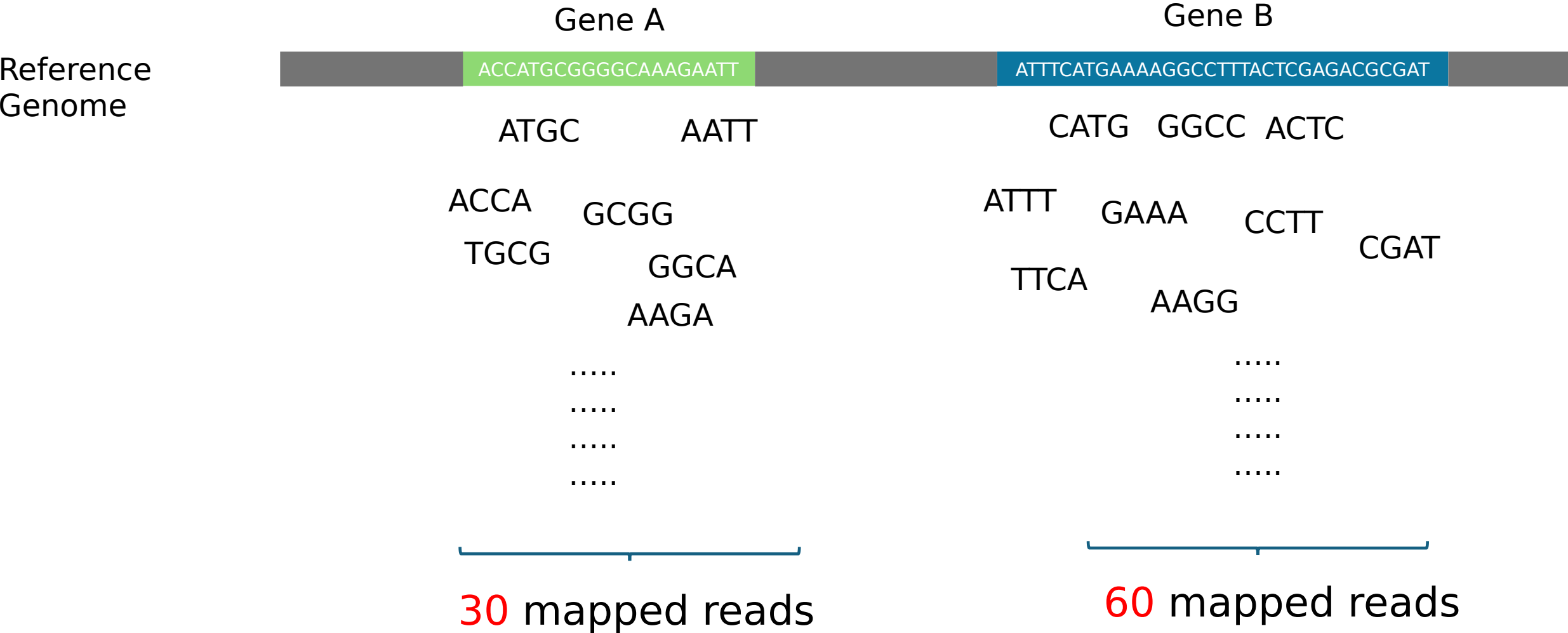
EM for RNA-seq (RSEM)

The multi-mapping problem and the EM solution — from BIOL419 Lecture 12.

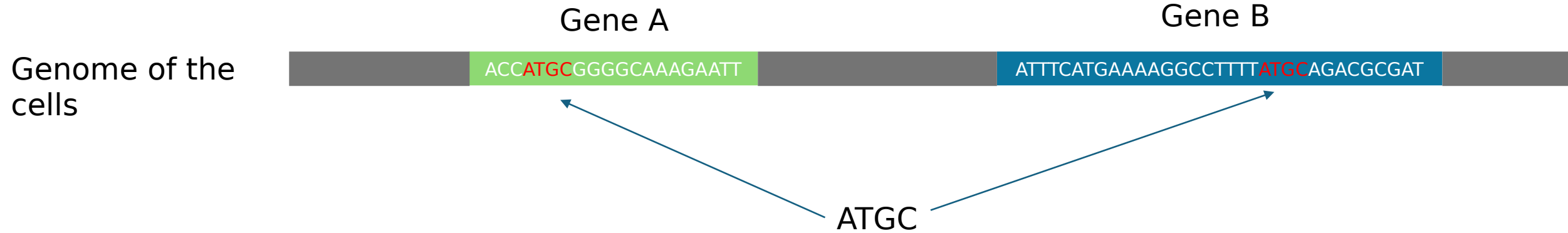
You: *RNA-seq data analysis is conceptually easy!*

Me: *Well... there are some complications.*

Simplified Read Mapping



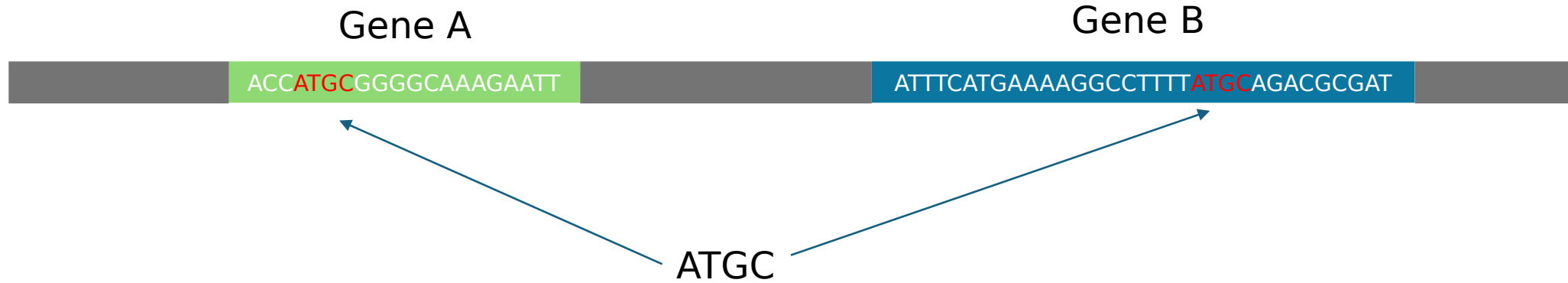
In reality



What should we do for this multi-mapping read?

In reality

Genome of the
cells

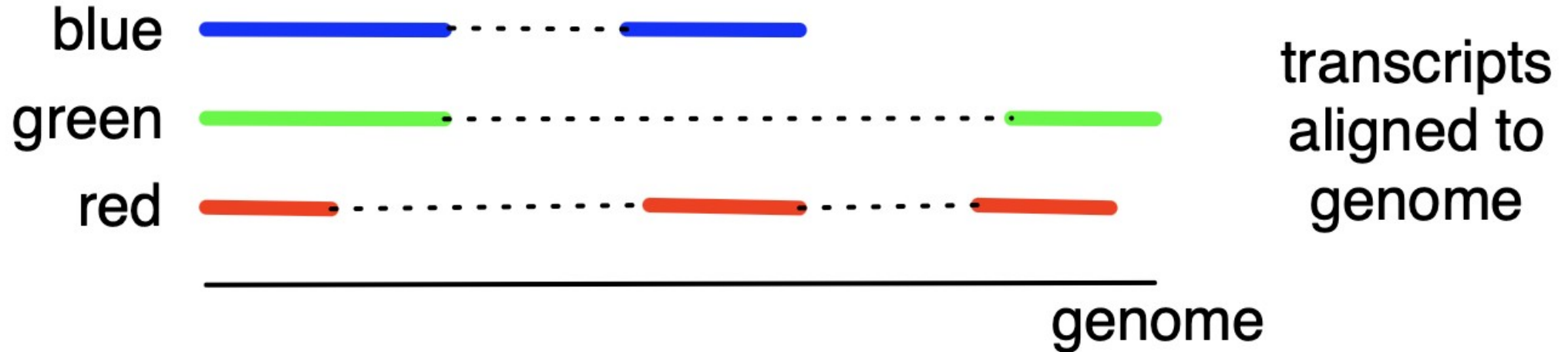


What should we do for this multi-mapping read?

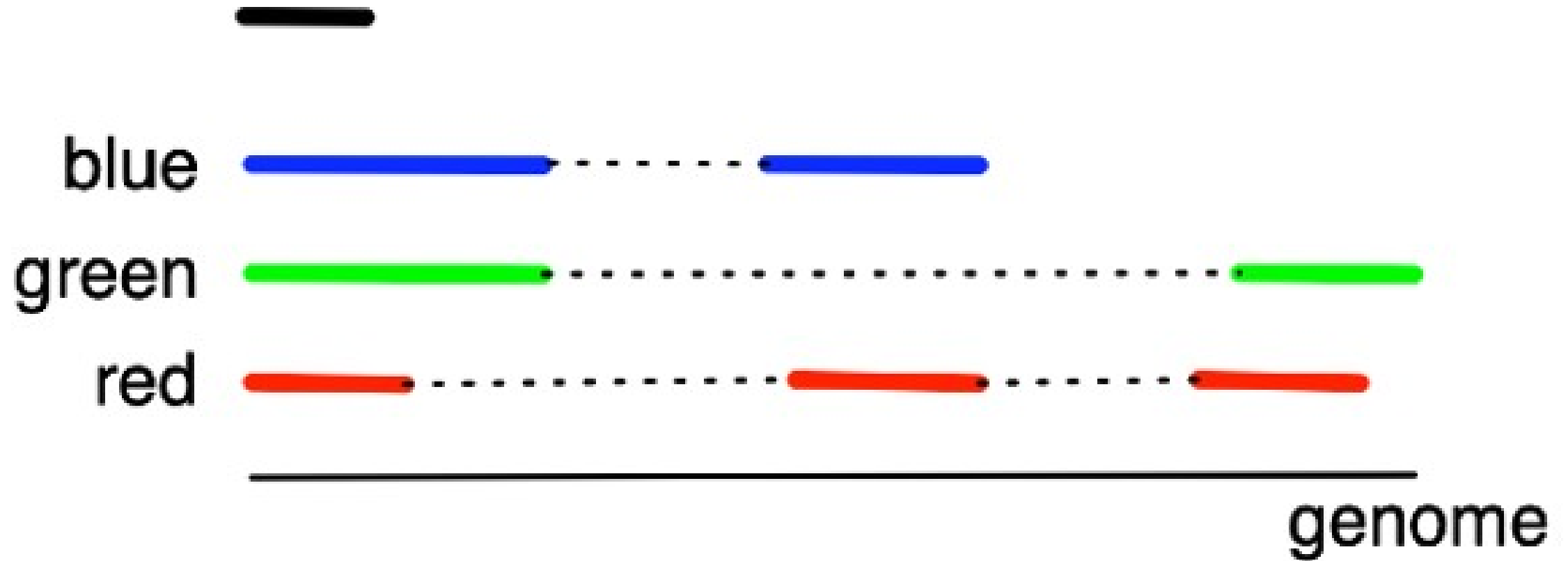
- 1: Drop the read, and pretend we have never seen it
- 2: Count 0.5 for **Gene A** and 0.5 for **Gene B**

A more challenging problem

Each human gene usually have multiple **transcripts**



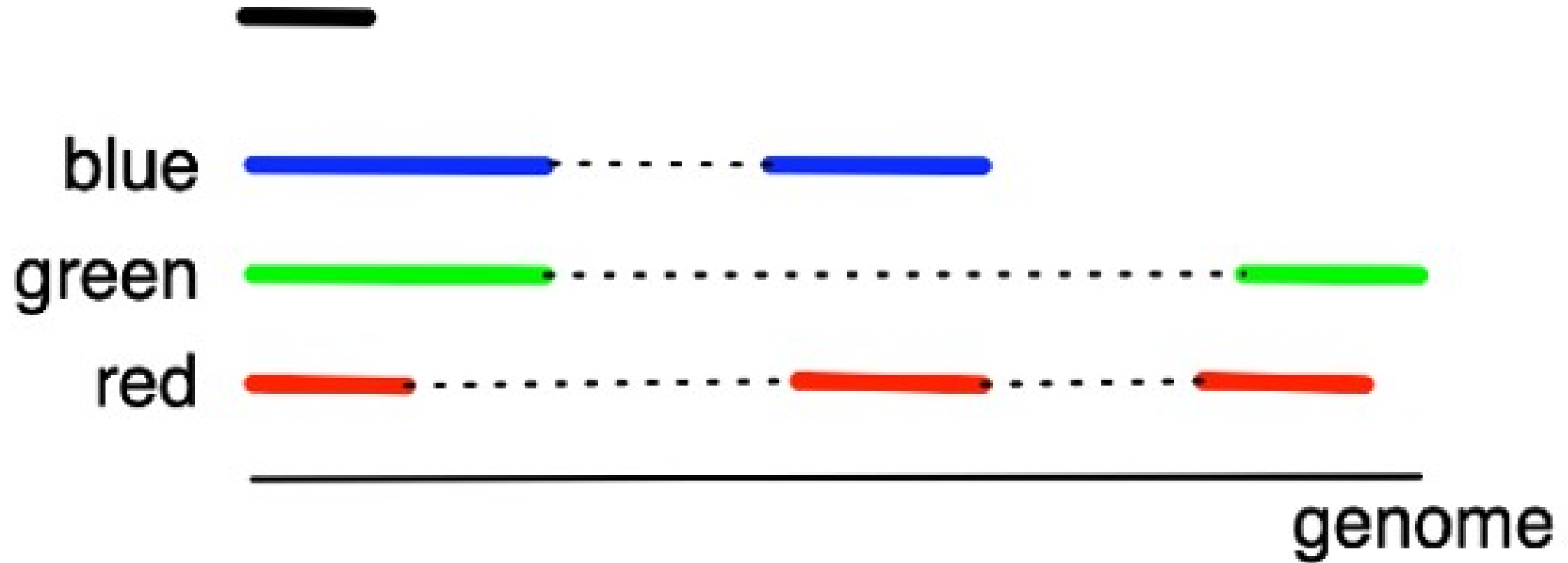
A more challenging problem



A more challenging problem

(1) Drop the read?

(2) Split the read to $\frac{1}{3}$, $\frac{1}{3}$, and $\frac{1}{3}$?

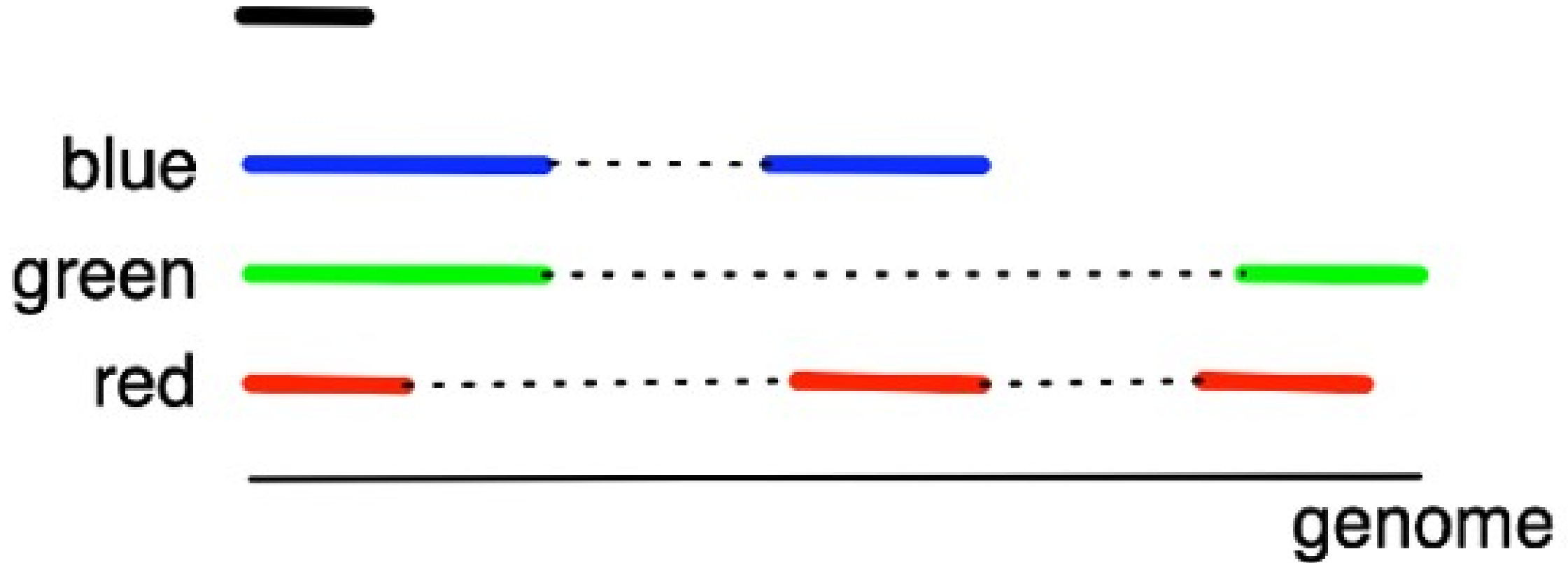


A more challenging problem

(1) Drop the read?

(2) Split the read to $\frac{1}{3}$, $\frac{1}{3}$, and $\frac{1}{3}$?

Or any better solution?



A more challenging problem

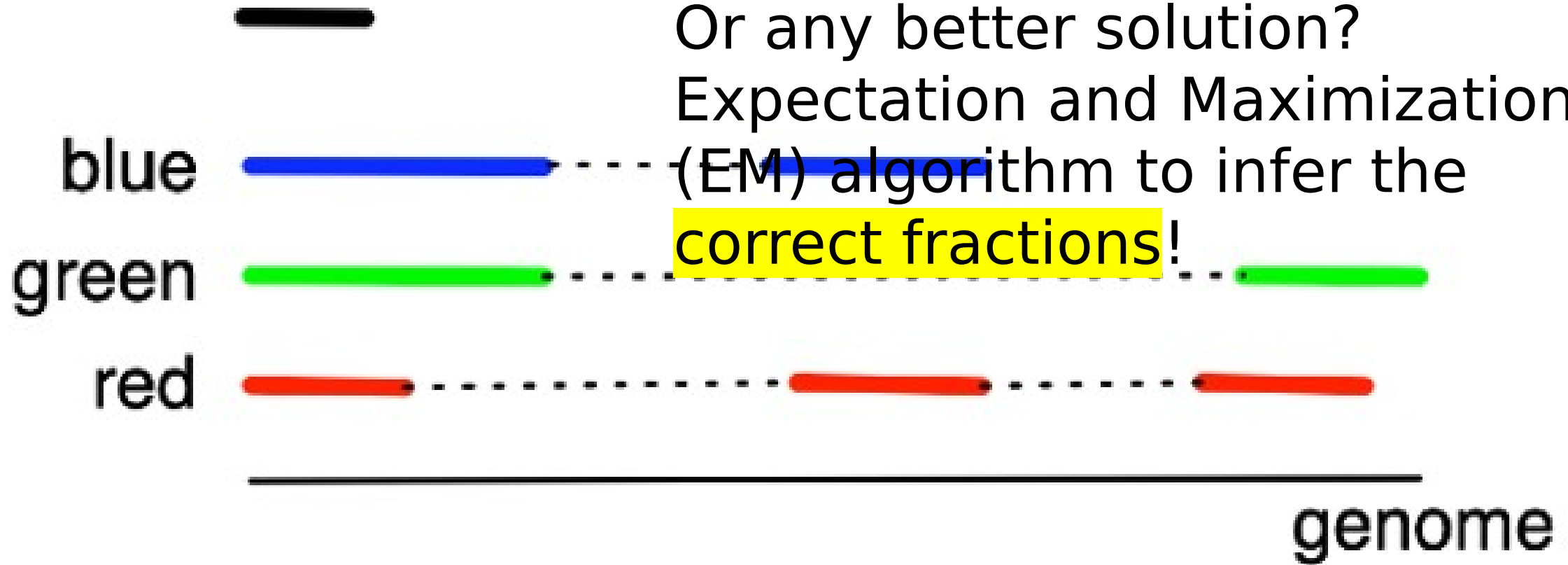
(1) Drop the read?

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Or any better solution?

Expectation and Maximization

(EM) algorithm to infer the
correct fractions!



Expectation-Maximization Algorithm for RNA-Seq

Expectation-Maximization Algorithm

E Step: Compute values of missing information z given an estimate of θ at time t .

$$q^t(z) = P(z|r, \theta^{(t)})$$

M Step: Update the estimates of θ based on new z values that maximize the data likelihood.

$$Q(\theta|\theta^{(t)}) = \sum_z q^t(z) \log[P(r, \hat{z}|\theta)]$$

$$\hat{\theta}^{(t+1)} = \operatorname{argmax}_{\theta} Q(\theta|\theta^{(t)})$$

$$\hat{\theta}^{(t+1)} = \operatorname{argmax}_{\theta} Q(\theta|\theta^{(t)})$$

$$\begin{aligned} \log[P(r|\theta)] &= \log\left(\sum_z P(r, z|\theta)\right) \\ &= \log\left[\sum_z q(z) \cdot \frac{P(r, z|\theta)}{q(z)}\right] = \log\left[E_{qz}\left(\frac{P(r, z|\theta)}{q(z)}\right)\right] \\ &\geq \sum_z q(z) \log\left[\frac{P(r, z|\theta)}{q(z)}\right] = E_{qz}\left[\log\left(\frac{P(r, z|\theta)}{q(z)}\right)\right] \end{aligned} \quad (2.1)$$

$$(2.2)$$

The penultimate step (from Equation 2.1 to 2.2) uses Jensen's inequality to get a lower bound on the log data likelihood, which is the quantity we want to maximize.

Definition 2.1 (Jenson's Inequality) Given a convex function f and random variable X ,

$$E[f(X)] \geq f(EX).$$

If f is strictly convex, then

$$E[f(X)] = f(EX) \iff X \text{ is constant.}$$

Jensen's inequality also holds for concave function f with the inequality reversed:

$$f(EX) \geq E[f(X)]$$

(Andrew 2018).

A strictly concave function is a function f where $f''(x) \leq 0$. In the terms of the EM algorithm, we conclude that $\log\left[E_{qz}\left(\frac{P(r, z|\theta)}{q(z)}\right)\right] \geq E_{qz}\left[\log\left(\frac{P(r, z|\theta)}{q(z)}\right)\right]$ from Equations 2.1 and 2.2. Here, $f(x) = \log(x)$, where $x = \frac{P(r, z|\theta)}{q(z)}$, is a strictly concave function because $f''(x) = -x^{-2} \leq 0$. Thus, $f(EX) \geq E[f(X)]$. Note that the inequality becomes an equality when $\frac{P(r, z|\theta)}{q(z)}$ is equal to a constant.

$$\begin{aligned} c &= \frac{P(r, z|\theta)}{q(z)} \\ q(z) &= \frac{P(r, z|\theta)}{c} \\ &= \frac{P(r, z|\theta)}{\sum_z P(r, z|\theta)} \text{ because } \sum_z q(z) = 1 \\ &= \frac{P(r, z|\theta)}{P(r|\theta)} \\ &= P(z|r, \theta) \end{aligned}$$

By setting $q(z) = P(z|r, \theta)$ (the posterior distribution of z over the data r and the current estimate of θ), we get an equality for the the log likelihood of the data, $\log(P(r|\theta))$. For the true value of θ (unknown), we can say:

$$\log(P(r|\theta)) = \sum_z q(z) \log\left(\frac{P(r, z|\theta)}{q(z)}\right) = \sum_z P(z|r, \theta) \log\left(\frac{P(r, z|\theta)}{P(z|r, \theta)}\right)$$

For the given estimate of θ at time t (θ^t), and the associated $q^t(z) = P(z|r, \theta^t)$ we can say:

$$\log(P(r|\theta^t)) = \sum_z q^t(z) \log\left(\frac{P(r, z|\theta^t)}{q^t(z)}\right) = \sum_z P(z|r, \theta^t) \log\left(\frac{P(r, z|\theta^t)}{P(z|r, \theta^t)}\right)$$

Here, we can get the parameter $\theta^{(t+1)}$ by maximizing the likelihood involving the previous estimate of the parameter. Because in this specific case, $q^t(z)$ is not defined as $P(z|r, \theta)$, but rather $P(z|r, \theta^t)$, the following is true due to Jenson's inequality (note there is no equality due to $q(z) \neq q^t(z)$):

$$\log(P(r|\theta)) \geq \sum_z q^t(z) \log\left(\frac{P(r, z|\theta^t)}{q^t(z)}\right) = \sum_z P(z|r, \theta^t) \log\left(\frac{P(r, z|\theta^t)}{P(z|r, \theta^t)}\right)$$

Further simplifying the log likelihood at a given estimate at time t gives us:

$$\begin{aligned} \log(P(r|\theta^t)) &= \sum_z q^t(z) \log\left(\frac{P(r, z|\theta^t)}{q^t(z)}\right) \\ &= \sum_z q^t(z) \log[P(r, z|\theta^t)] - \sum_z q^t(z) \log(q^t(z)) \end{aligned}$$

Note that the second term is independent of θ . Because we want to maximize the likelihood with respect to θ , we only need to maximize

$$Q(\theta|\theta^{(t)}) = E_{q^t(z)}[\log(P(r, z|\theta^t))] = \sum_z q^t(z) \log(P(r, z|\theta^t))$$

where $\theta^{(t)}$ is the current estimate of θ . We now iterate between the following two steps of the algorithm, where each iteration never decreases the likelihood of the data under the estimated parameters (Dempster, Laird, and Rubin 1977).

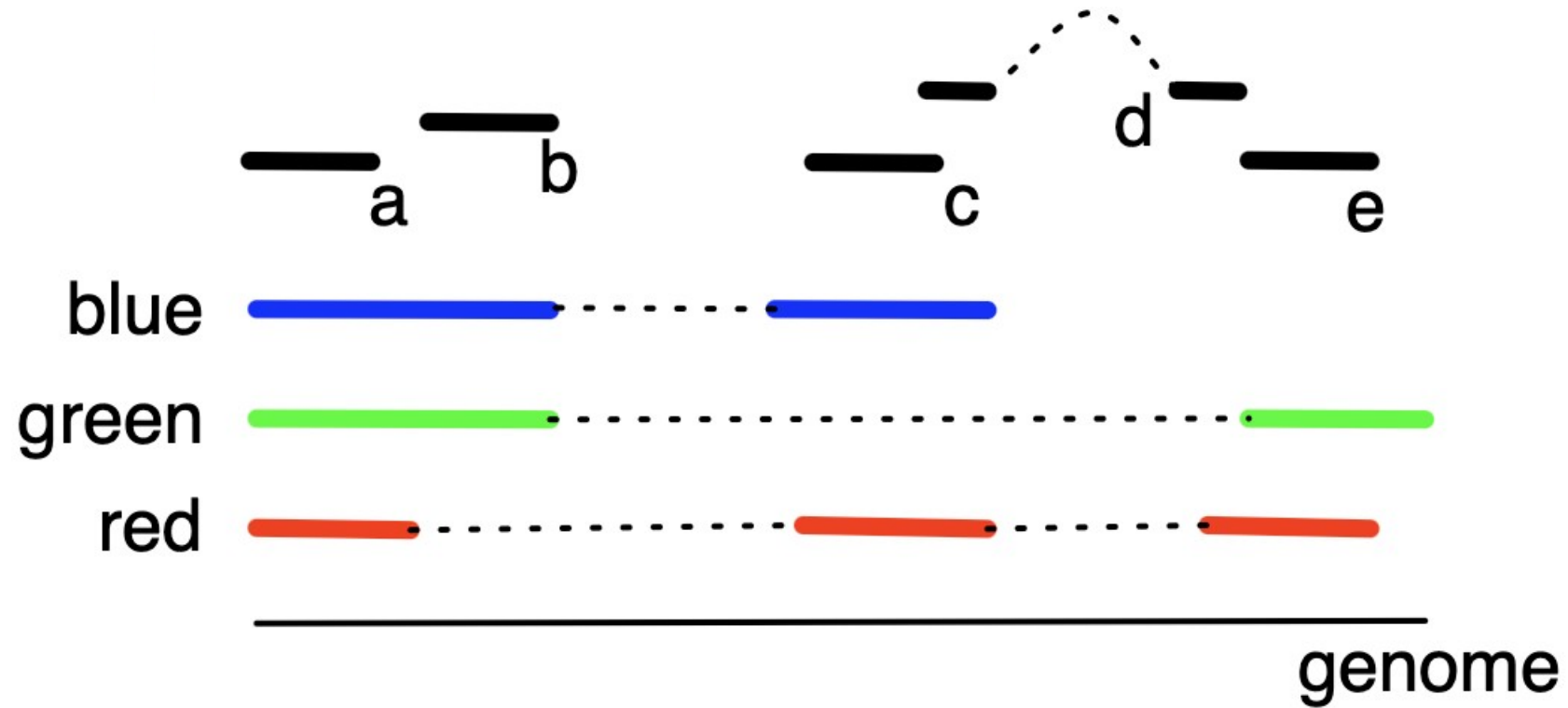
1. E step: $Q(\theta|\theta^{(t)})$ is the expected value of the log likelihood function of θ with respect to the current conditional distribution of z given an estimate of θ and data r .

$$q(z) = P(z|r, \theta^{(t)})$$

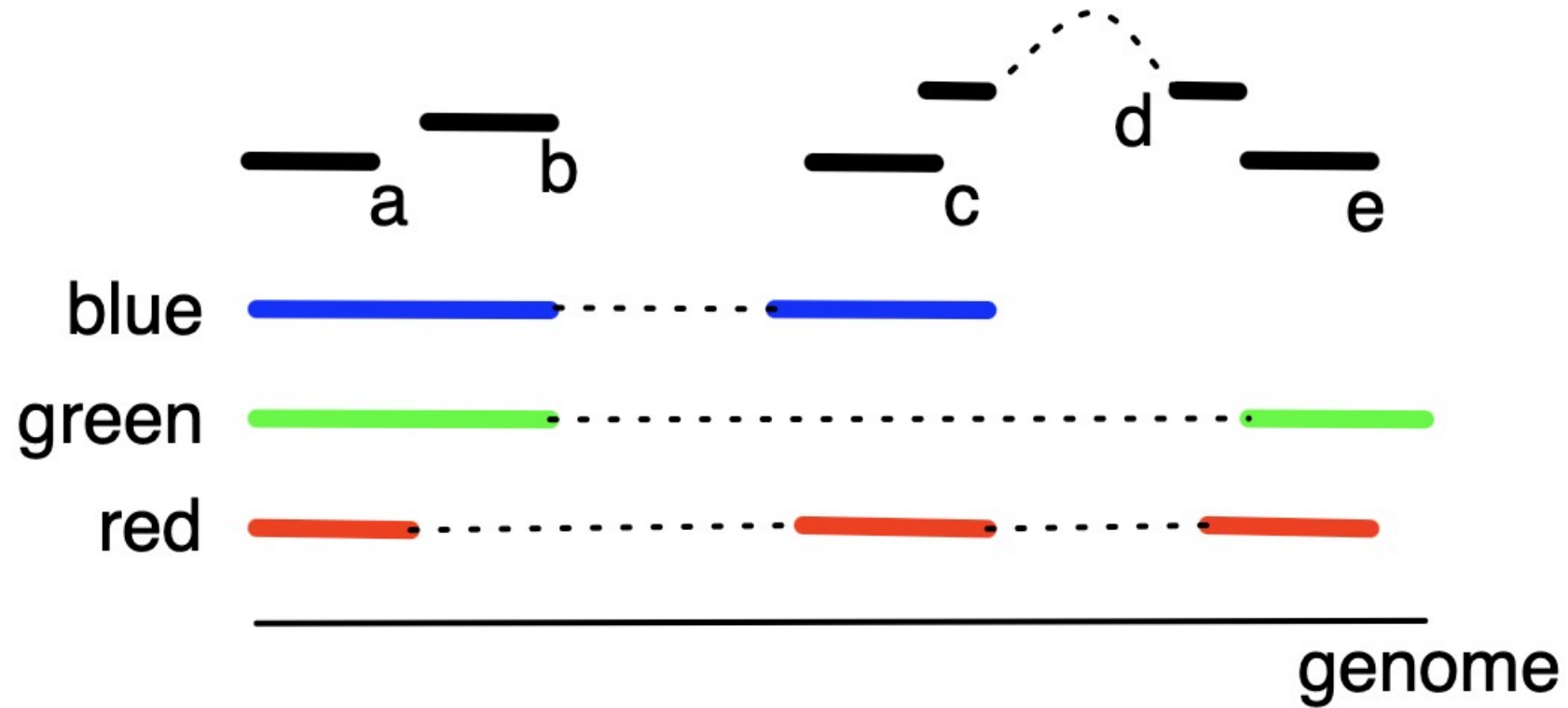
$$Q(\theta|\theta^{(t)}) = \sum_z q(z) \log(P(r, z|\theta))$$

$$Q(\theta|\theta^{(t)}) = \sum_z P(z|r, \theta^{(t)}) \log(P(r, z|\theta))$$

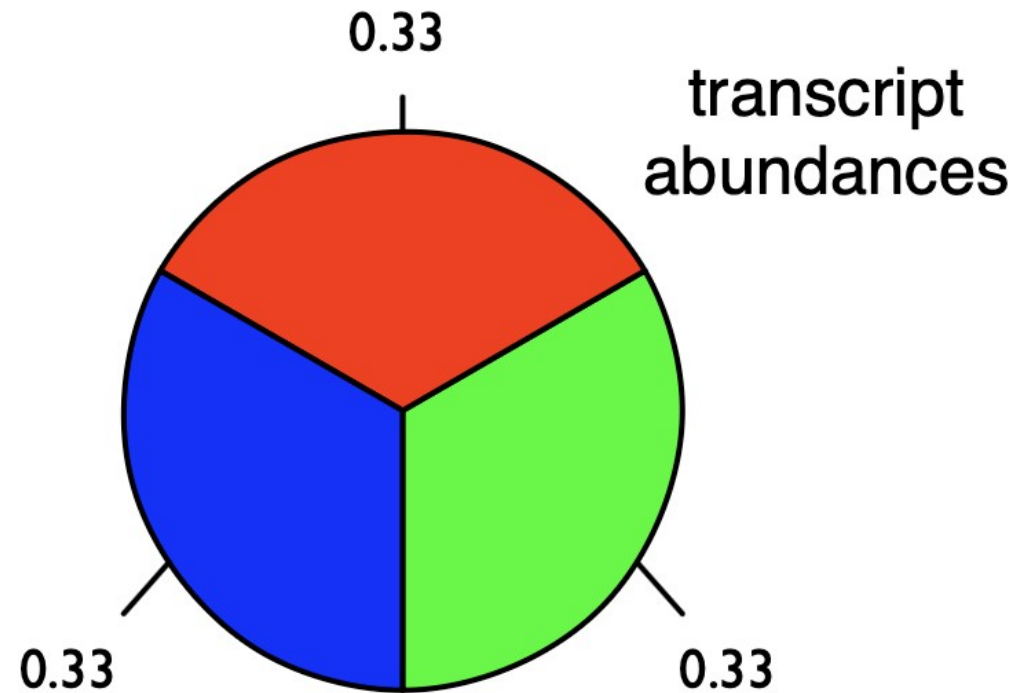
Let's learn by example and use just you intuition!



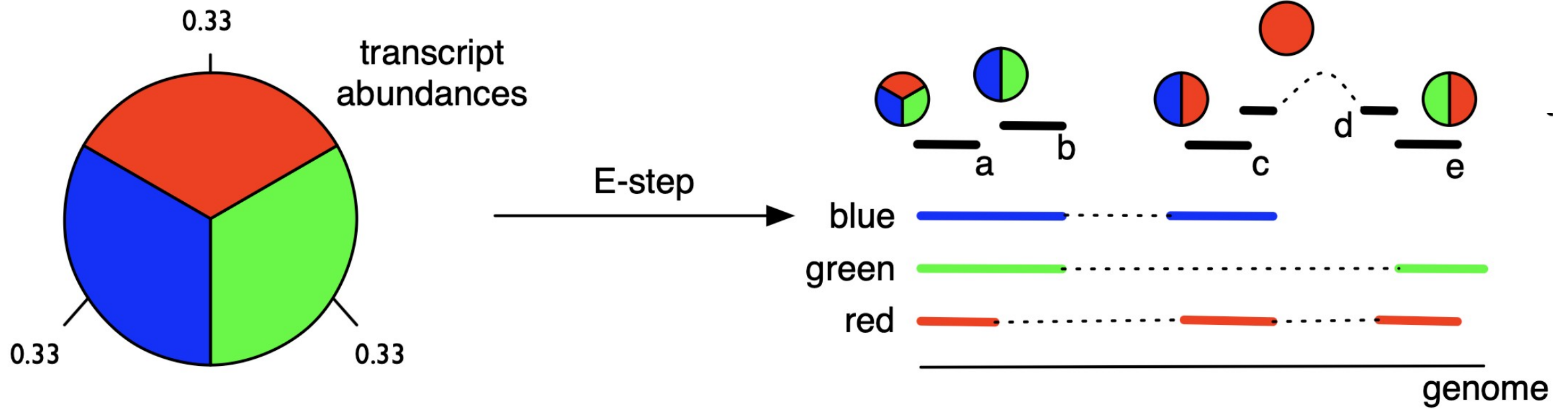
E-M Principle: reads are more likely to be mapped to transcripts with higher abundances



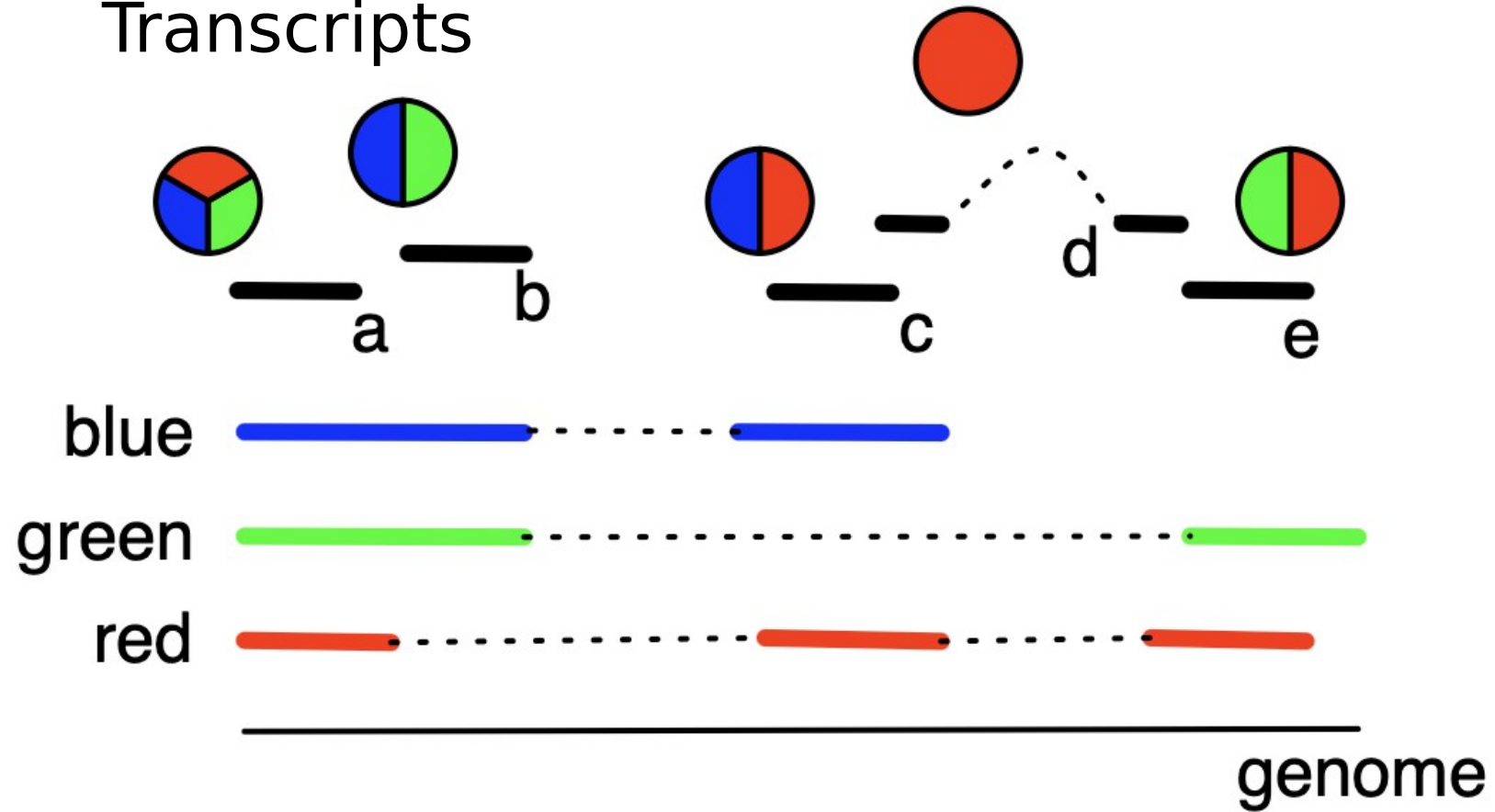
Initiation: All the transcripts have the same abundances



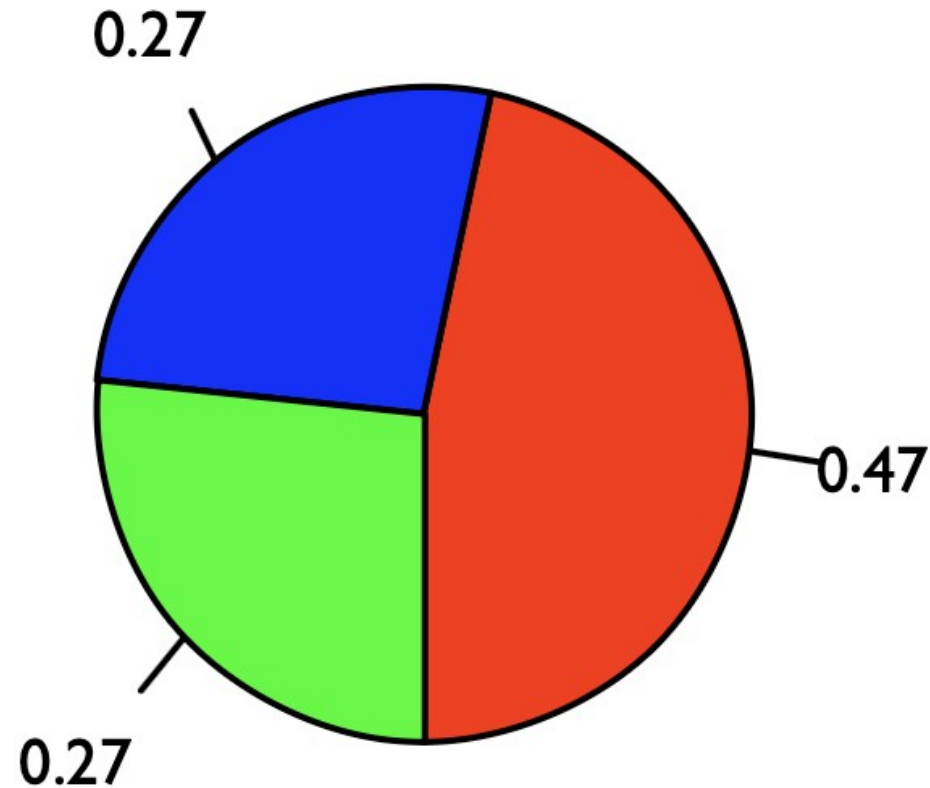
E-Step: Update the Classification of Reads



M-Step: Estimate the Abundances of Transcripts

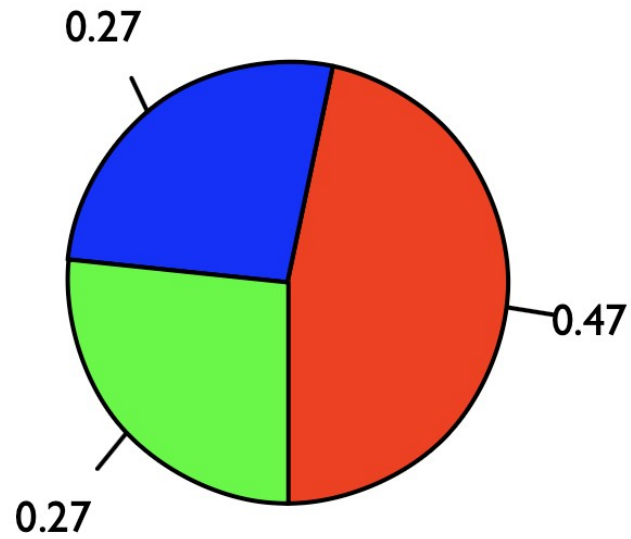


M-Step results in an **Updated Transcript Abundances**

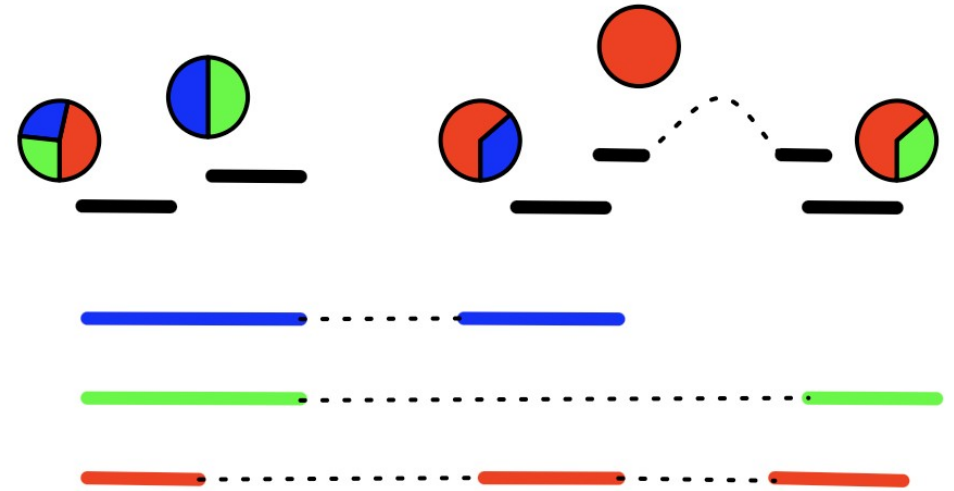


E-Step: Update the Classification of

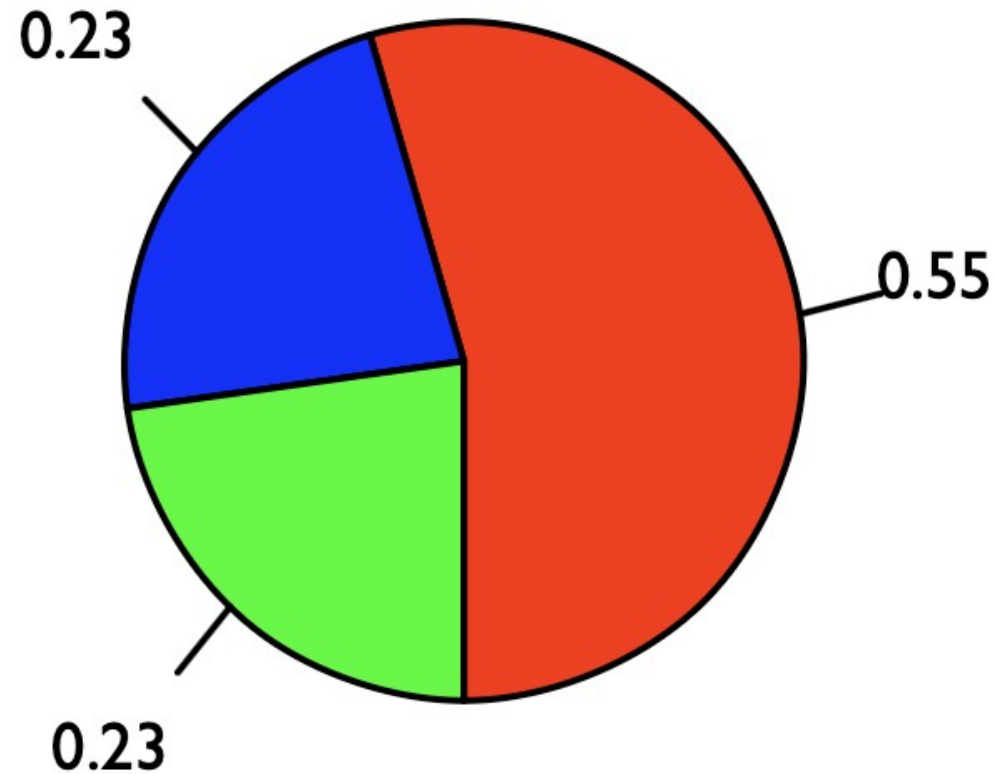
Nodes

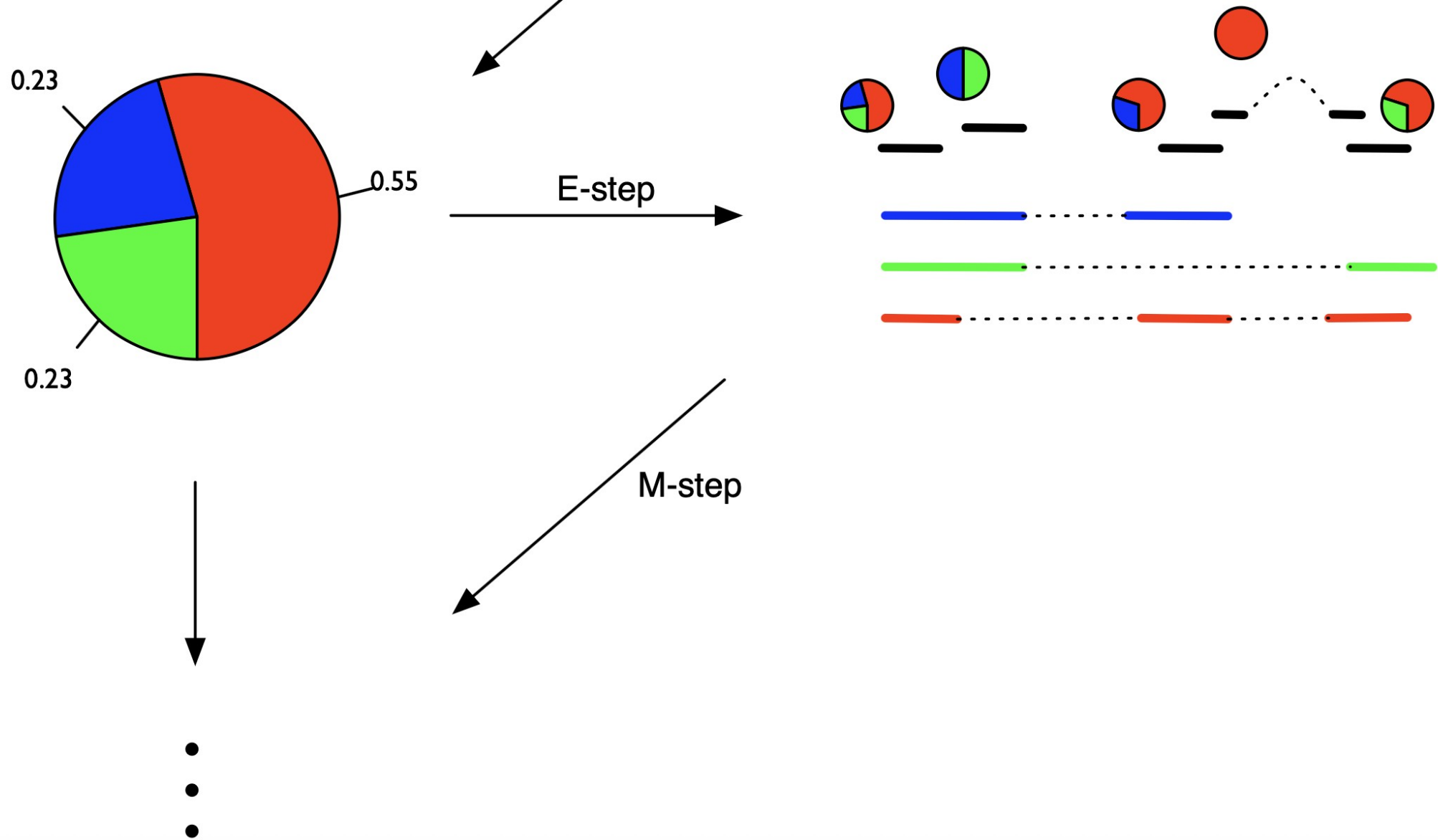


E-step



M-Step results in an **Updated Transcript Abundances**





Do E-Step and M-Step iteratively, until transcript abundances converge

Output of EM-based Software for RNA-seq

(1) Isoform (Transcript) Expression Levels

- Unit is TMP or RPKM or FPKM

(2) Gene expression levels

- e.g. Gene A = IsoformA_1 + IsoformA_2 +

The same EM in ten lines

Two transcripts. 6 reads unique to A, 2 unique to B, 4 compatible with both.

Python

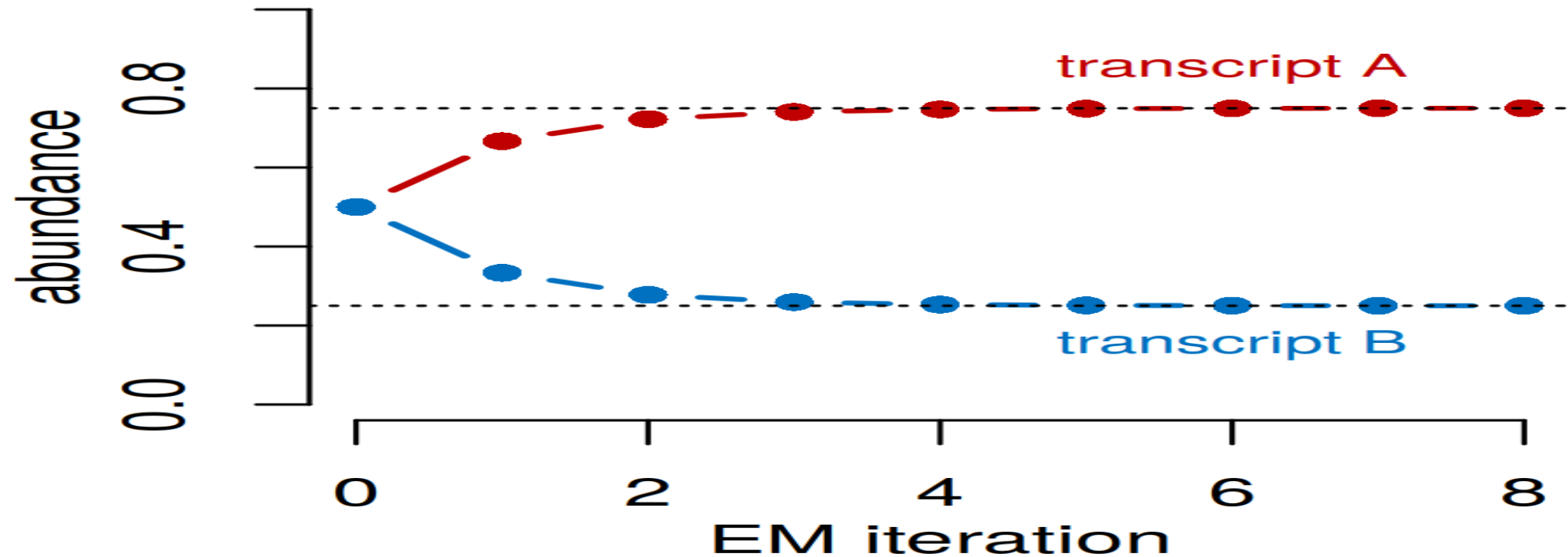
```
nA, nB, nShared = 6, 2, 4
theta = [0.5, 0.5]          # initiation
for it in range(8):
    # E-step: split shared reads
    eA = nA + nShared * theta[0] / sum(theta)
    eB = nB + nShared * theta[1] / sum(theta)
    # M-step: new abundances
    theta = [eA / (eA + eB), eB / (eA + eB)]
    print(it + 1, [round(t, 3) for t in theta])
# 1 [0.667, 0.333] ... 6 [0.75, 0.25]
```

R

```
nA <- 6; nB <- 2; nShared <- 4
theta <- c(0.5, 0.5)        # initiation
for (it in 1:8) {
    # E-step: split shared reads
    eA <- nA + nShared * theta[1] / sum(theta)
    eB <- nB + nShared * theta[2] / sum(theta)
    # M-step: new abundances
    theta <- c(eA, eB) / (eA + eB)
    cat(it, round(theta, 3), "\n")
}
# 1 0.667 0.333 ... 6 0.75 0.25
```

Naive 50/50 splitting gives $A = 0.667$. EM converges to 0.75: the more abundant transcript claims more of the shared reads, self-consistently.

Convergence in six iterations



Dotted lines: the fixed point 0.75 / 0.25. Each E-M round moves toward it and never overshoots.

RSEM does this for every isoform of every gene with every multi-mapping read — millions of hidden labels at once.

RSEM output is the E-step's last assignment

- Your BIOL419 Lab 4 file `sample_rsem.genes.results` — now `chr22_expression.tsv` on the course site.
- **expected_count = 317.84 for IGLC2**: a fractional count, because shared reads were split by probability, not by coin flip.
- TPM then normalizes expected counts by effective length and to one million — the L7 formula.
- **ACTBP15, a pseudogene, has 650 expected reads**: with only chr22 in the reference, β -actin reads had nowhere else to go. EM cannot fix a missing reference.

EM resolves ambiguity among the choices it is given. Give it the whole genome, or it will confidently assign reads to the wrong place.

Part 6

DESeq2: the concepts, in order

Normalize → model the spread → borrow strength → test → pay the burden.

Step 1 — Make samples comparable (size factors)

- Libraries differ in depth: 15.7 M to 31.8 M reads in airway. A gene with 100 reads in a 15 M library and 200 in a 30 M library did not change.
- **Why not divide by total reads?** A few giant genes (immunoglobulins, mitochondrial genes) can be a third of all reads; if they change, every other gene appears to change the other way.
- **Median of ratios:** for each gene, compare each sample to a reference (the gene's geometric mean); the sample's size factor is the median of those ratios across genes — the typical gene decides, not the giants.
- Airway size factors: 1.02, 0.90, 1.18, 0.67, 1.17, 1.39, 0.92, 0.95.

Step 2 — Model the spread (negative binomial per gene)

- Part 1: donors add spread beyond Poisson. DESeq2 models each gene's counts as NB with its own dispersion α .
- **The design formula $\sim \text{cell} + \text{dex}$** says: each donor has its own baseline (pairing!), and dex shifts every donor by the same log fold change.
- This is the $y \sim \text{group}$ formula from yesterday's boxplot and t.test, with one more term. The fold change you report is the coefficient of dex.
- A t-test on log counts ignored both the NB spread and the pairing. That is why it found so little.

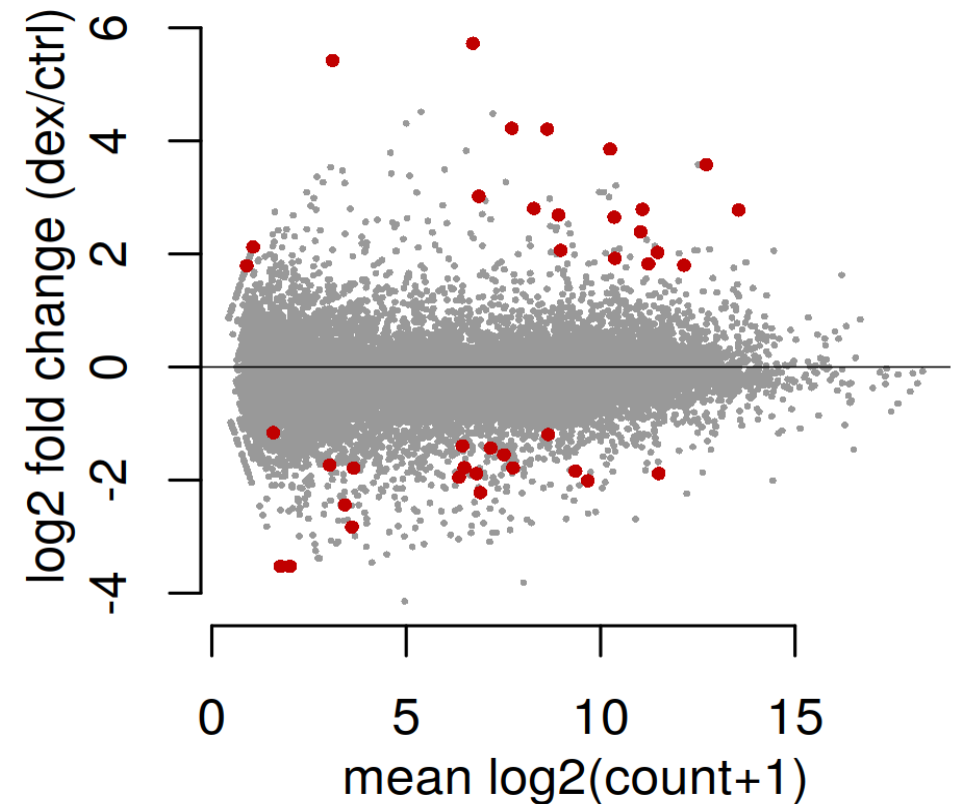
Step 3 — Borrow strength (why 4 vs 4 can work)

- With four values you cannot estimate a spread reliably: some genes look perfectly consistent by luck and would produce absurdly small p-values.
- **The trick:** genes with similar mean counts have similar dispersions. Fit the trend of α vs mean across all 19,271 genes, then pull each gene's own α toward the trend.
- A gene with lots of information keeps its own estimate; a gene with little information borrows from the crowd.
- The same idea (empirical Bayes shrinkage) is in limma (2004), edgeR (2010), DESeq2 (2014). It is what makes small experiments honest.

You have 4 replicates per gene but 19,271 genes per replicate. Use them.

Step 4 — Test, then pay the burden

- **Wald test:** fold change $\div$ its uncertainty $\rightarrow$ one p-value per gene (the coin logic again).
- **Benjamini-Hochberg:** padj column = the q-value from Part 3. Filter on padj, never on p.
- **Shrink fold changes (lfcShrink):** a 3-vs-0 gene should not top your ranking; low-count fold changes are pulled toward zero.
- Then look: an MA plot (fold change vs mean) shows whether the changes are spread across expression levels or piled up where counts are tiny.



MA plot: red = genes surviving BH at 5 % (t-test version)

Reading a DESeq2 results table

- **baseMean** — average normalized count; below ~10 be suspicious.
- **log2FoldChange** — the effect size; +1 = doubled, -1 = halved.
- **lfcSE, stat** — uncertainty of the fold change, and their ratio (the Wald statistic).
- **pvalue** — one coin question per gene.
- **padj** — after BH. The column your list is built from.
- On the airway data, DESeq2 with ~ cell + dex finds several thousand genes at padj < 0.1 (DESeq2 vignette). The t-test found 41. The difference is the model, not leniency.

Three lines on your laptop: `DESeqDataSetFromMatrix(cts, meta, ~ cell + dex) → DESeq() → results()`.
Setup page has the script.

Reading a DESeq2 result

YOUR TURN A gene has $\log_2\text{FoldChange} = 4.1$, $\text{baseMean} = 3$, $\text{padj} = 0.04$. Do you put it in the paper's headline figure?

- A. Yes — $\text{padj} < 0.05$ and a 17-fold change
- B. No — $\text{baseMean} = 3$ means a handful of reads; the fold change is unstable
- C. Only after `lfcShrink` and a look at the individual counts

ANSWER C. Significance says the change is unlikely to be zero; it does not say the size is reliable. Low counts → shrink, then look at the eight numbers.

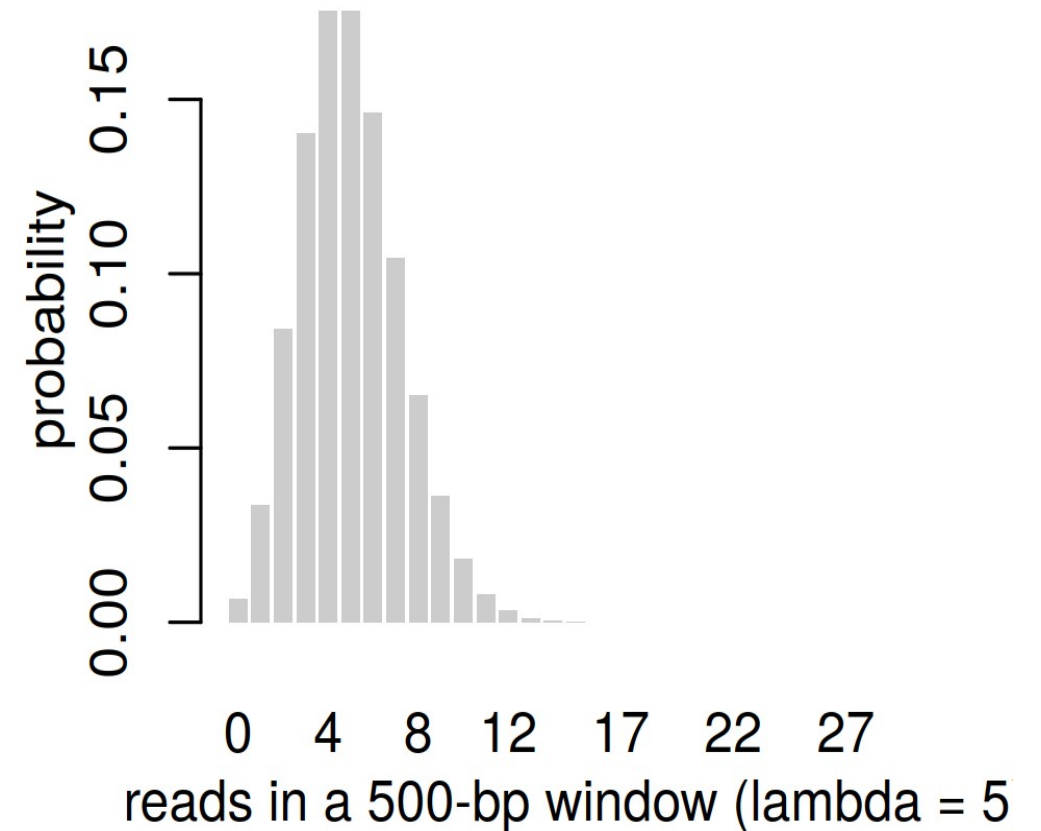
Part 7

ChIP-seq: is this peak real, and against what?

Rice on a grid meets the control question — from BIOL419 Lecture 13 and Xu et al., NAR 2021.

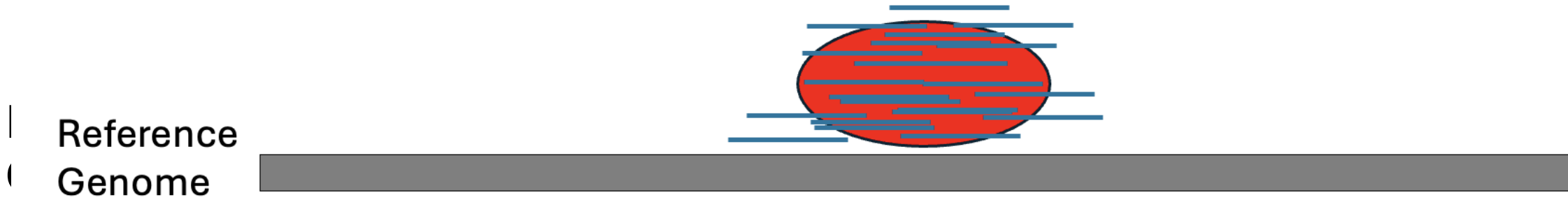
A peak is a Poisson question

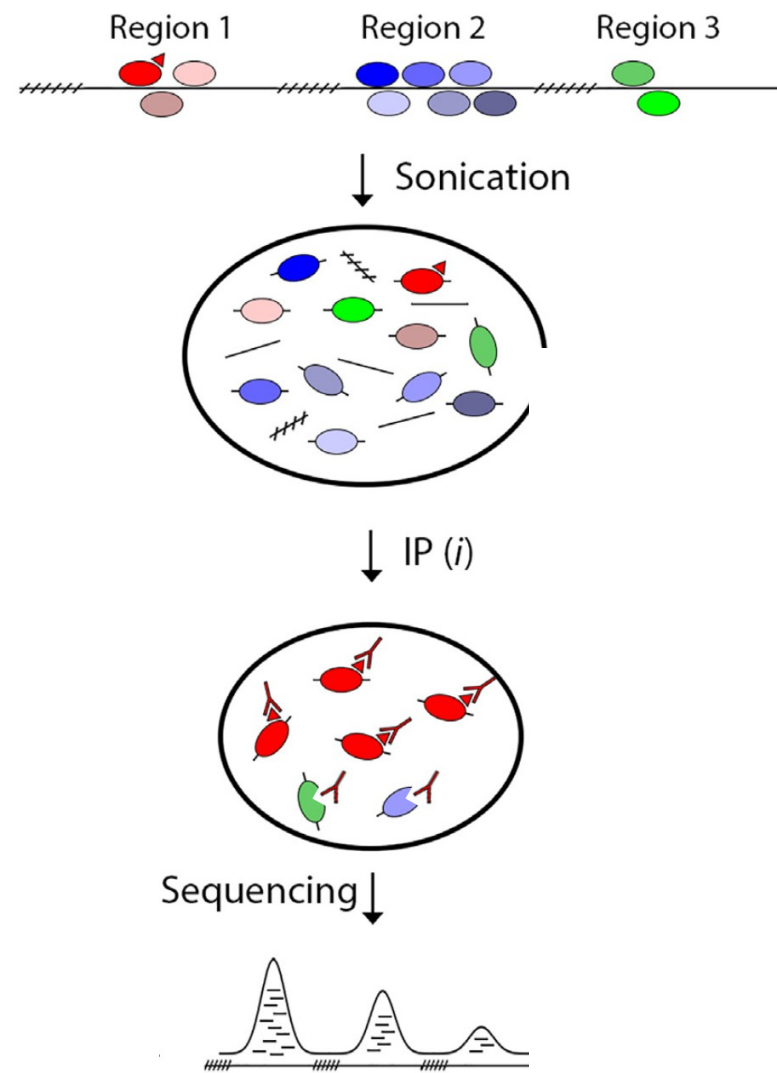
- Reads scattered at random over the genome give a background of, say, 5 reads per 500-bp window (the rice grid).
- A window with 25 reads: how often does $\text{Poisson}(5)$ produce 25 or more? 1.6×10^{-10} . That is the peak's p-value.
- **The whole difficulty is the background.** Open chromatin shears more easily, GC-rich regions sequence differently, some regions attract every antibody. Uniform scattering is false.
- So the background must be measured with a control experiment, window by window (MACS: a local λ). Which control?

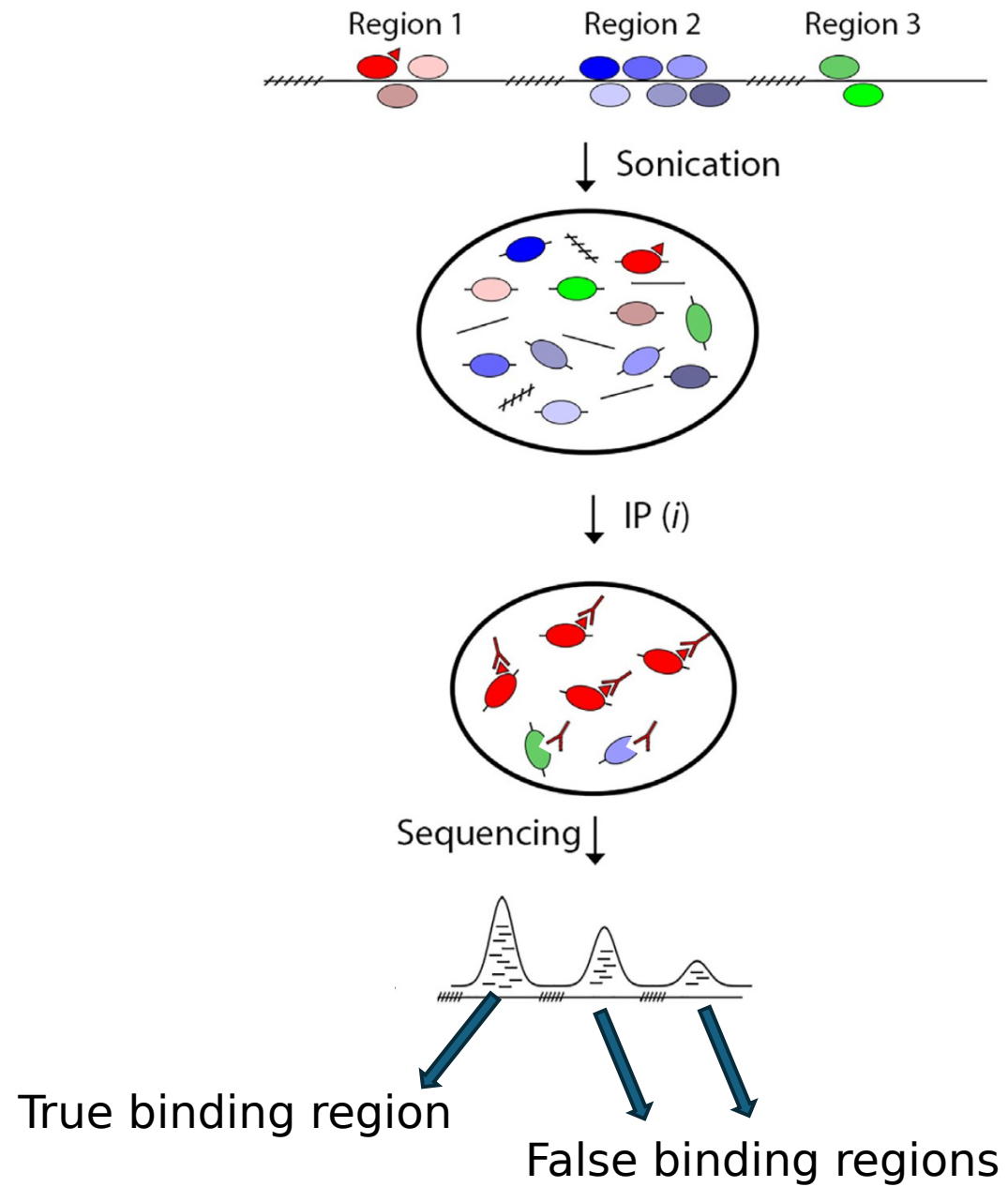


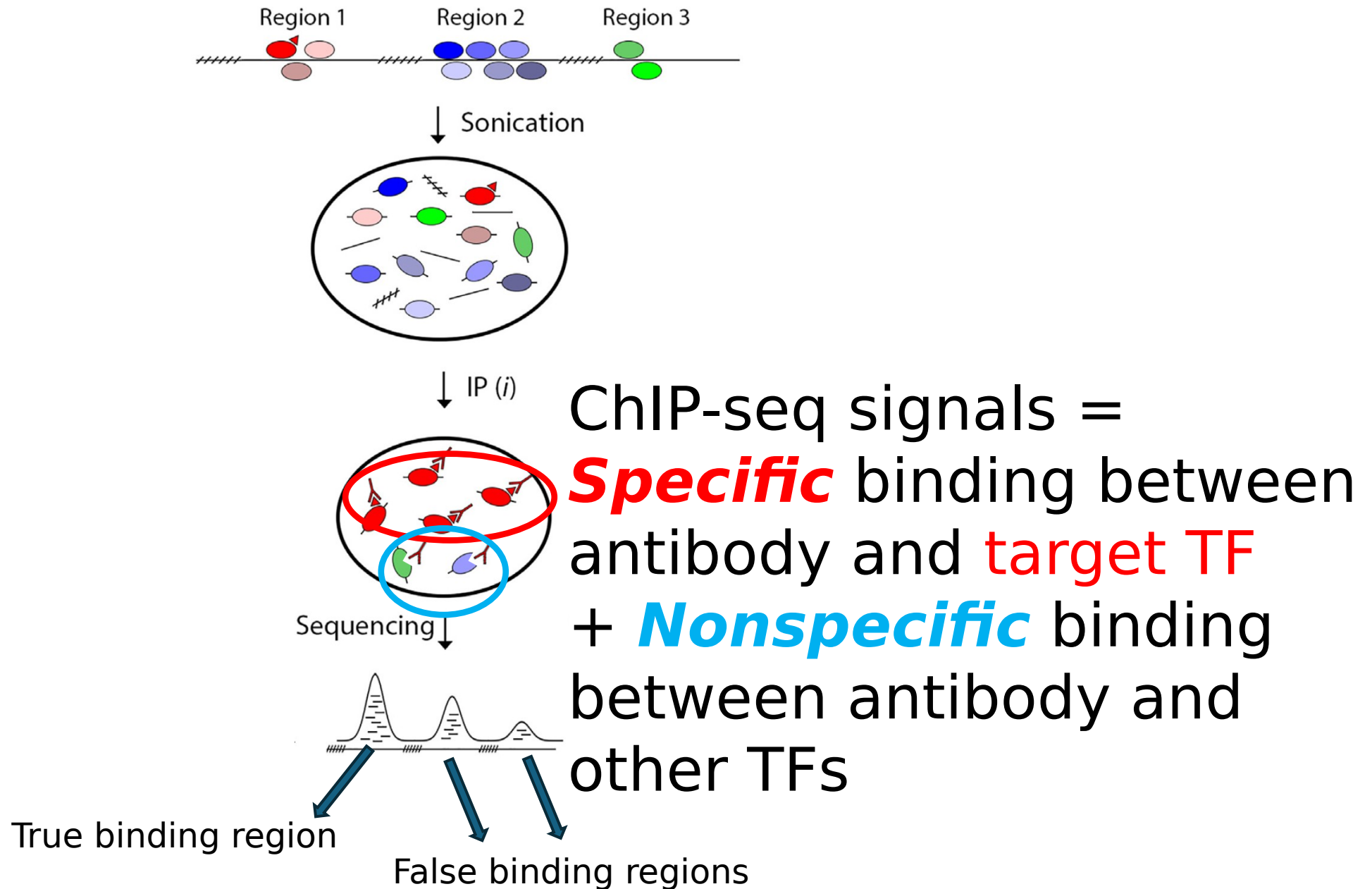
Poisson(5): the red tail (≥ 25 reads) is the peak

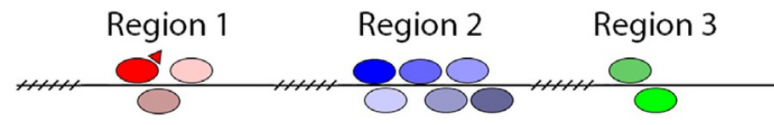
Chromatin Immuno-Precipitation followed by Sequencing (**ChIP-Seq**)



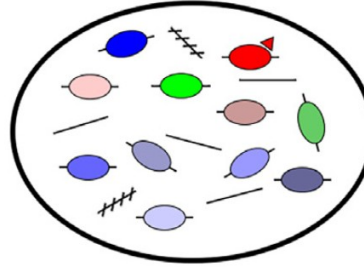




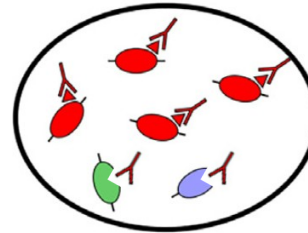




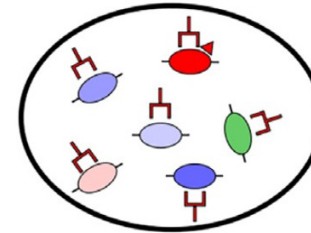
↓ Sonication



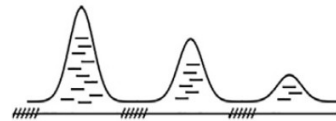
↓ IP (*i*)



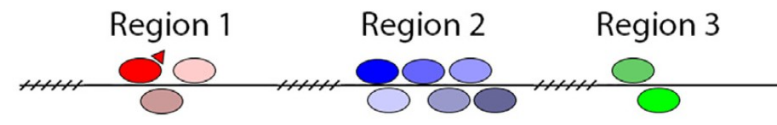
↘ Mock IP (*m*)



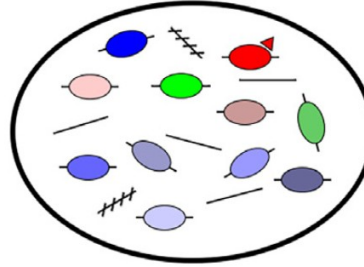
Sequencing ↓



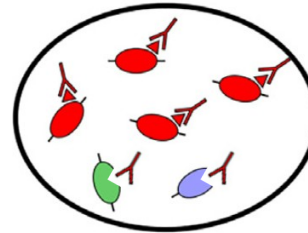
Use control
experiment to
mimic nonspecific
binding



↓ Sonication

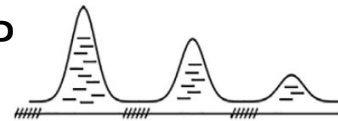


↓ IP (*i*)

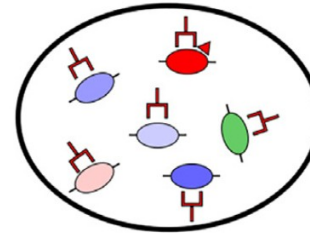


Sequencing ↓

Profile IP

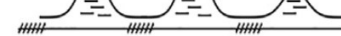


↘ Mock IP (*m*)

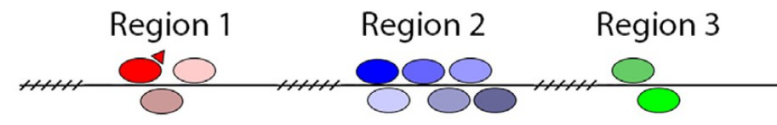


Sequencing ↓

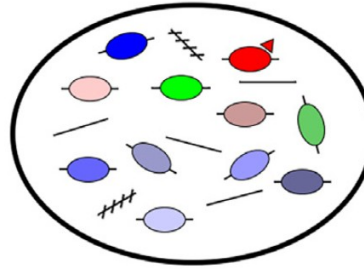
Profile Mock IP



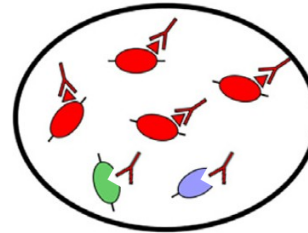
Use control experiment
to mimic nonspecific
binding



↓ Sonication

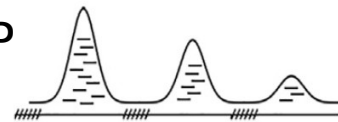


↓ IP (*i*)

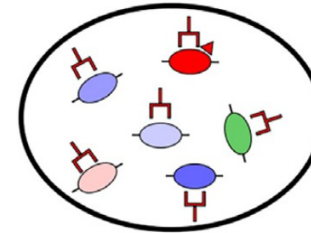


Sequencing ↓

Profile IP

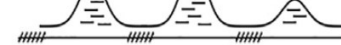


↘ Mock IP (*m*)



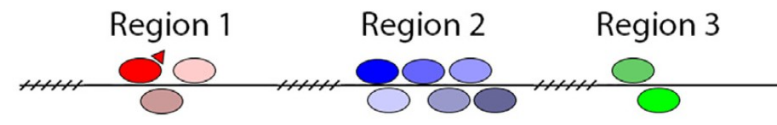
Sequencing ↓

Profile Mock IP

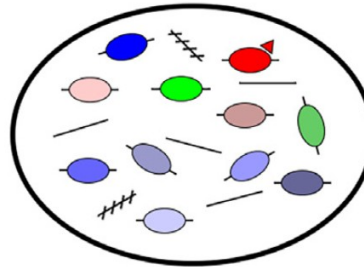


Use control experiment
to mimic nonspecific
binding

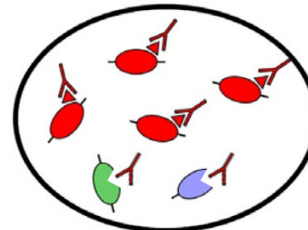
Profile of IP minus Profile of
Mock IP =



↓ Sonication

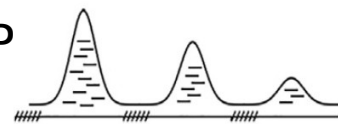


↓ IP (*i*)

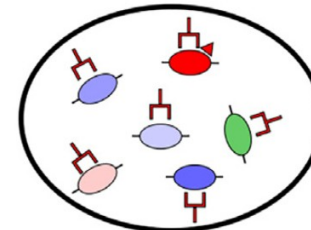


Sequencing ↓

Profile IP

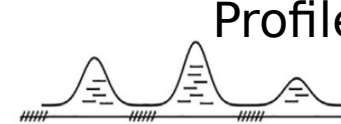


↘ Mock IP (*m*)

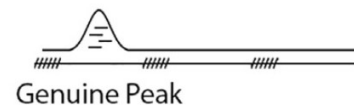


Sequencing ↓

Profile Mock IP



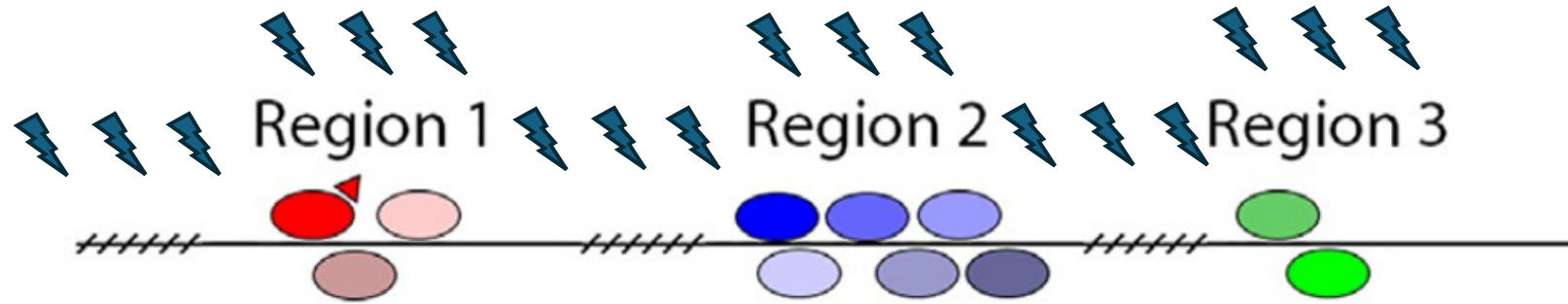
Peak Caller ↘ ✓



Use control experiment to mimic nonspecific binding

Profile of IP minus Profile of Mock IP =

Sonication Bias



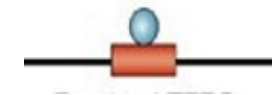
//// Closed Chromatin

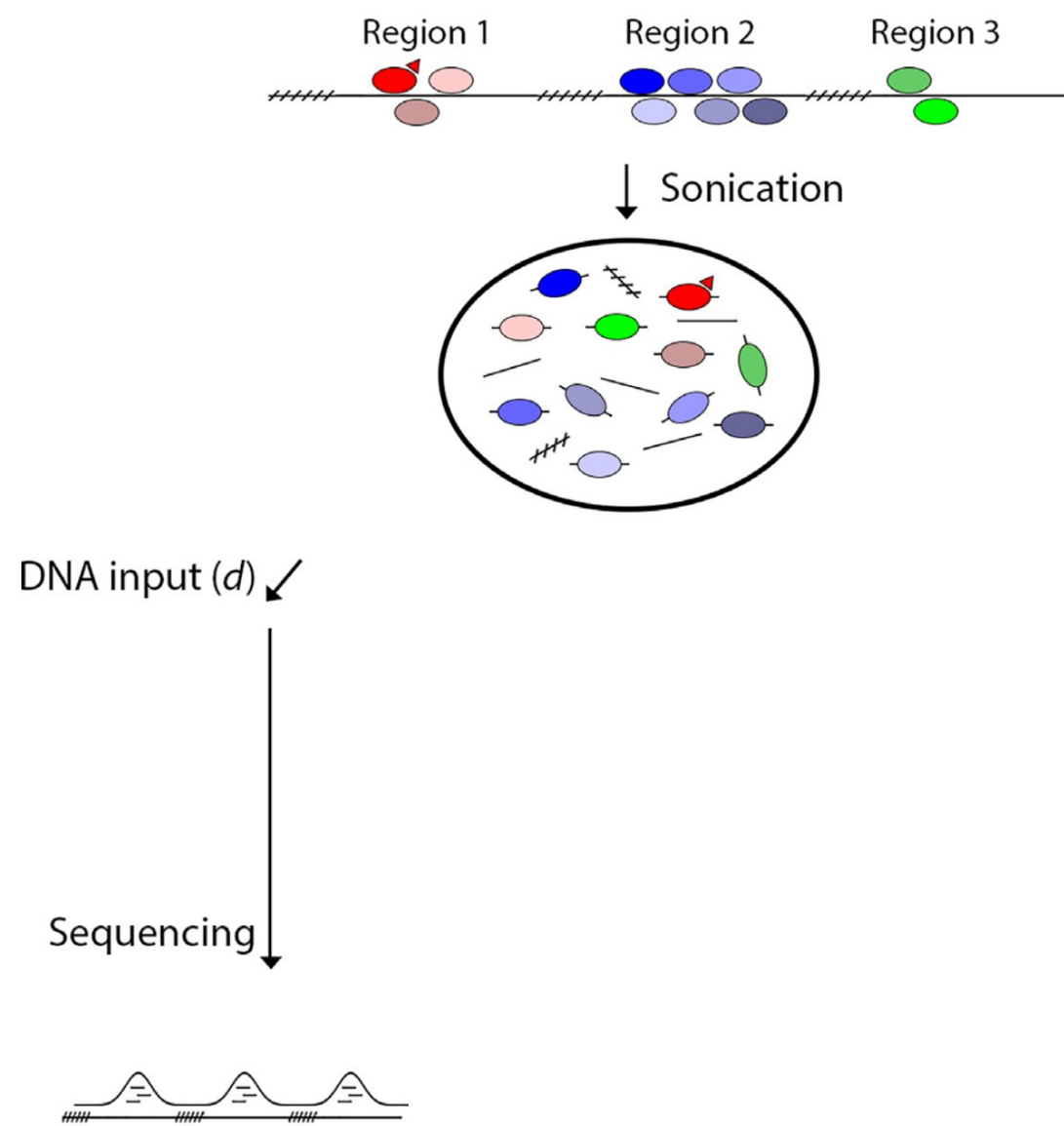
DNA densely packed with histones

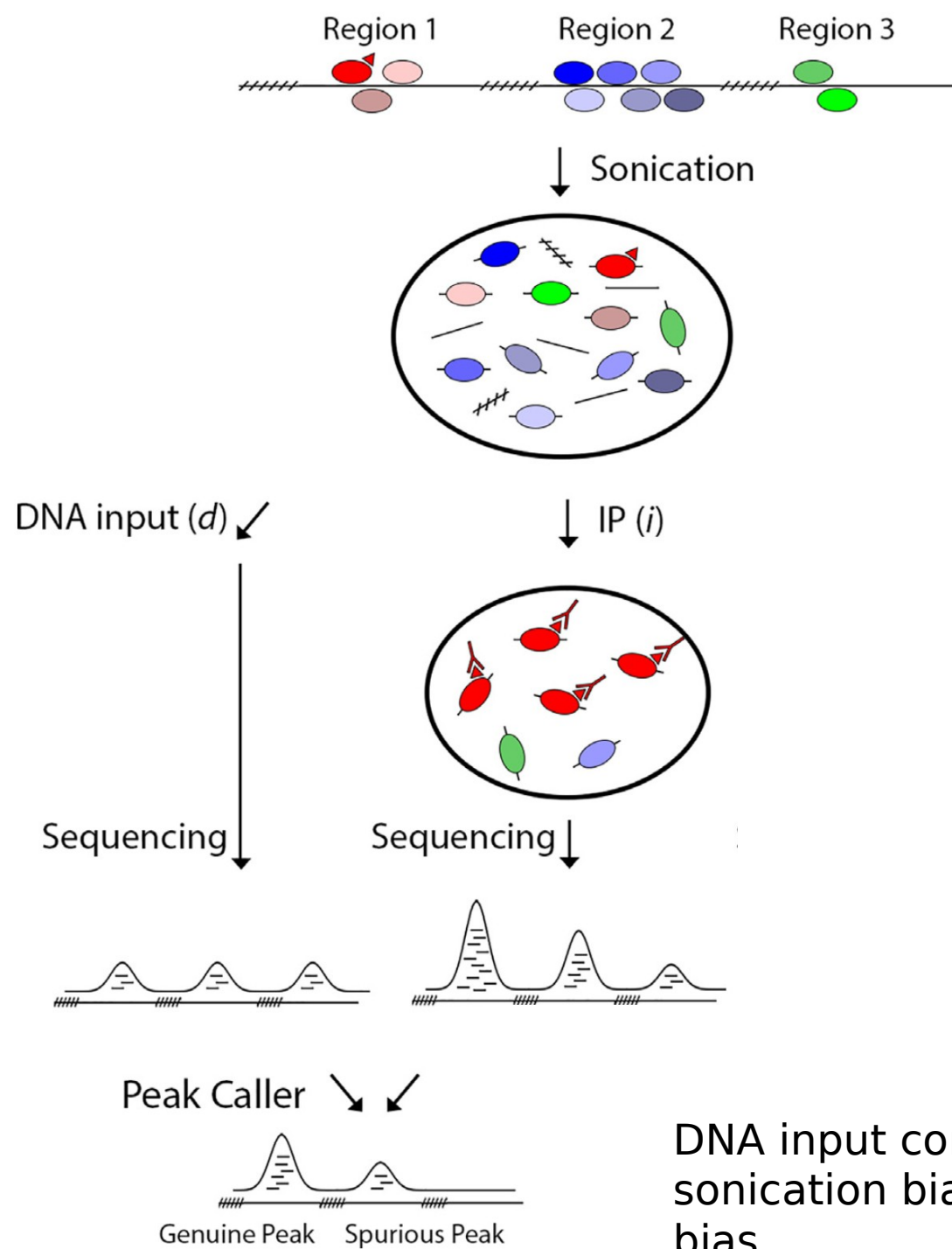


— Open Chromatin

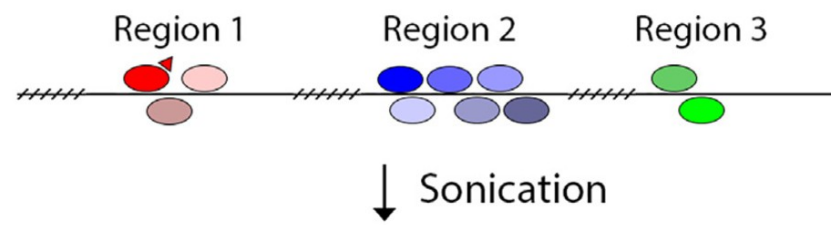
Naked DNA





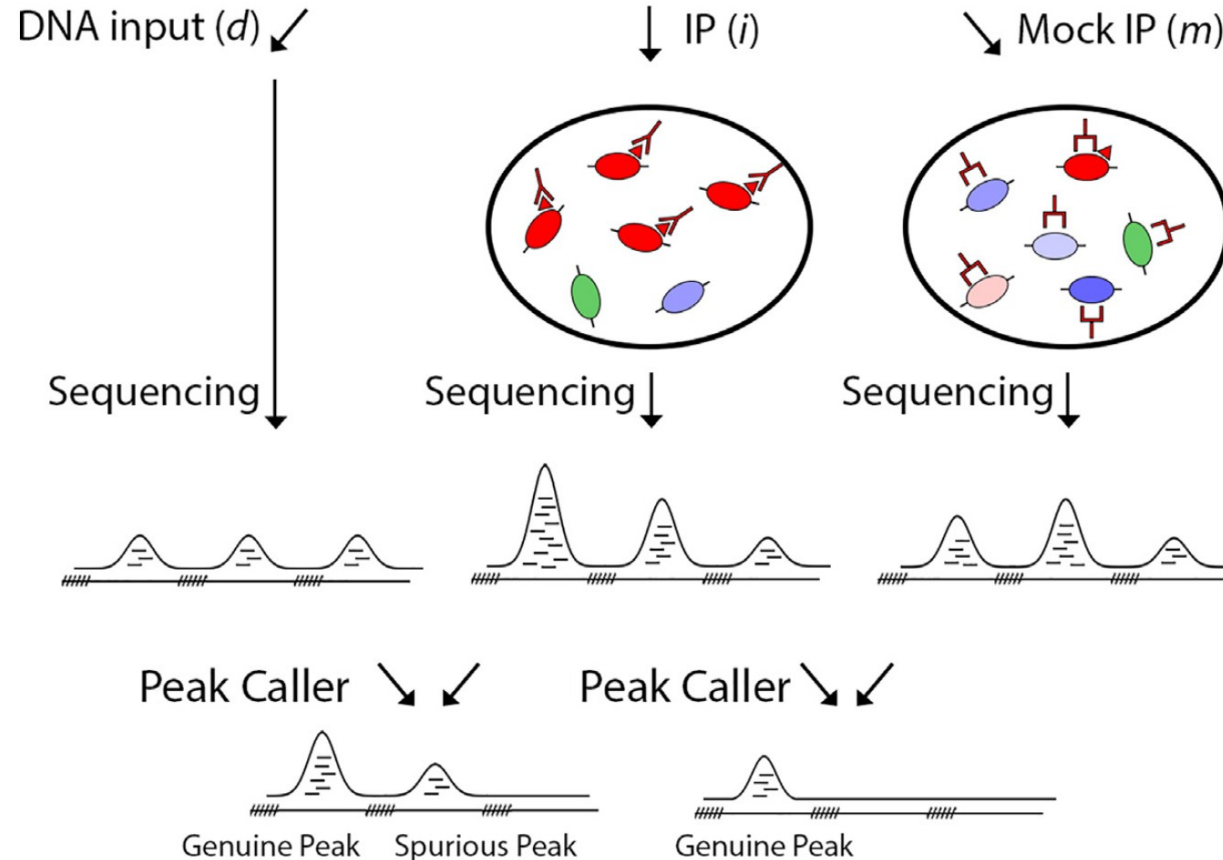


DNA input corrects only
sonication bias but binding
bias

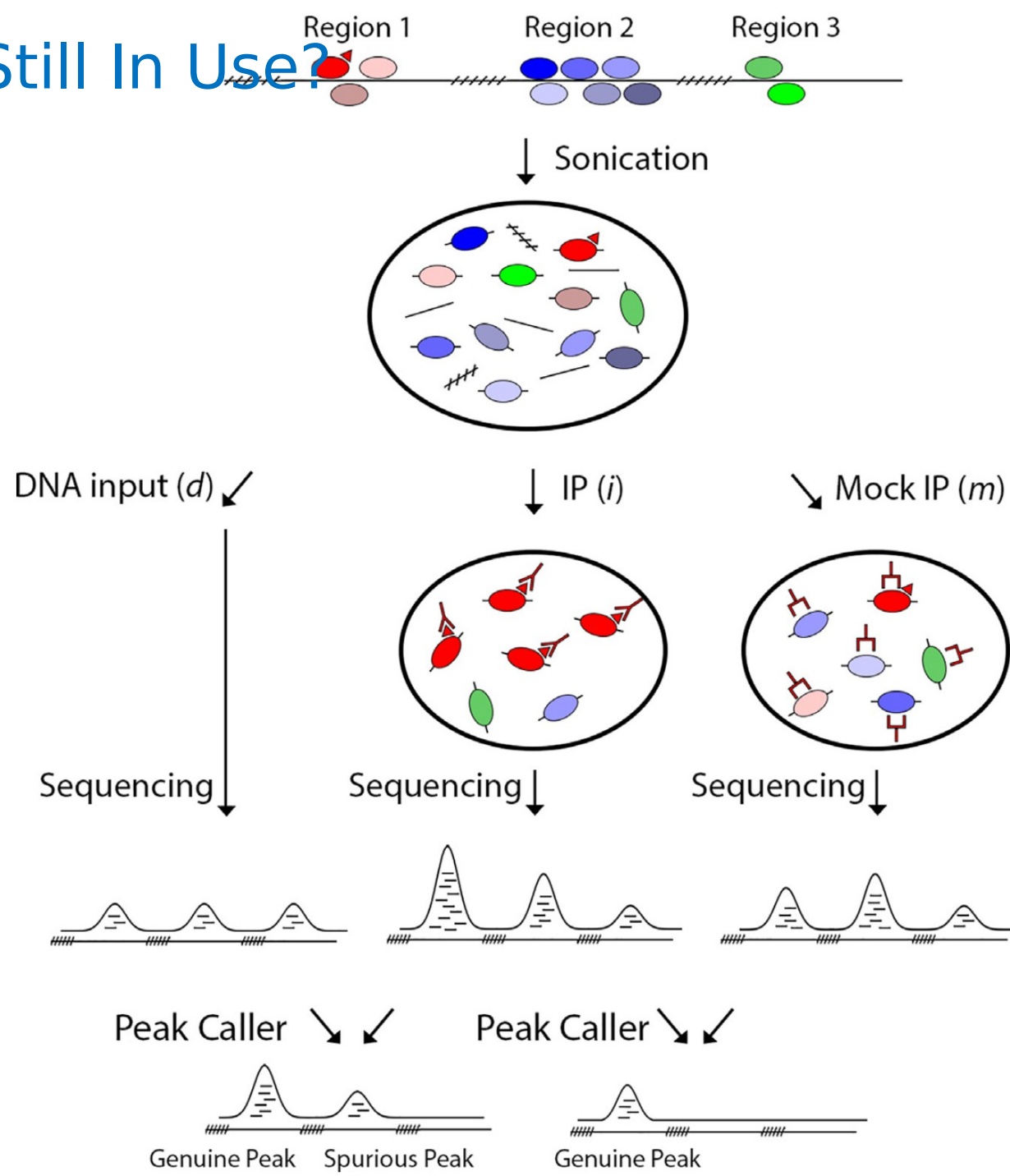


DNA input controls only
sonication bias

Mock IP controls both
non-specific interaction
and sonication bias



Why DNA Input Still In Use?



Maximizing the Advantage of both DNA input and Mock IP

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To mock or not: a comprehensive comparison of mock IP and DNA input for ChIP-seq

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$$P(\theta_1 > 0.5 \ \& \ \theta_1 > \theta_2) = \int_{\theta_2=0}^{\theta_1} \int_{\theta_1=0.5}^1 \text{Beta}(\theta_1 \mid n_i + 1, n_d + 1) \\ \text{Beta}(\theta_2 \mid n_{m'} + 1, n_{d'} + 1) d\theta_1 d\theta_2$$

King the Advantage of both DNA input and Mock IP

$$P(\theta_1 > 0.5 \ \& \ \theta_1 > \theta_2) = \int_{\theta_2=0}^{\theta_1} \int_{\theta_1=0.5}^1 \text{Beta}(\theta_1 \mid n_i + 1, n_d + 1) \\ \text{Beta}(\theta_2 \mid n_{m'} + 1, n_{d'} + 1) d\theta_1 d\theta_2$$

Maximizing the Advantage of both DNA input and Mock IP

$$P(\theta_1 > 0.5 \ \& \ \theta_1 > \theta_2) = \int_{\theta_2=0}^{\theta_1} \int_{\theta_1=0.5}^1 \text{Beta}(\theta_1 \mid n_i + 1, n_d + 1) \\ \text{Beta}(\theta_2 \mid n_{m'} + 1, n_{d'} + 1) d\theta_1 d\theta_2$$

To mock or not: what the data said (Xu et al., NAR 2021)

- modERN datasets with matched IP, DNA input and mock IP: 113 human (6 cell lines), 317 worm, 182 fly experiments.
- **Mock IP removes far more spurious sites than input:** ~10 % of input-called peaks in human cell lines, ~60 % in worm and ~88 % in fly (whole animals, many cell types).
- **Peaks surviving mock IP are more often real:** motif enrichment in top peaks rises 18–37 %.
- But mock IP libraries are shallow and noisy (very little DNA); used alone they lose sensitivity.
- **Recommendation: use both.** Call peaks against input (sensitivity), then score each peak by IP-vs-mock with a posterior probability (Part 4's Bayes).

The posterior, in plain words

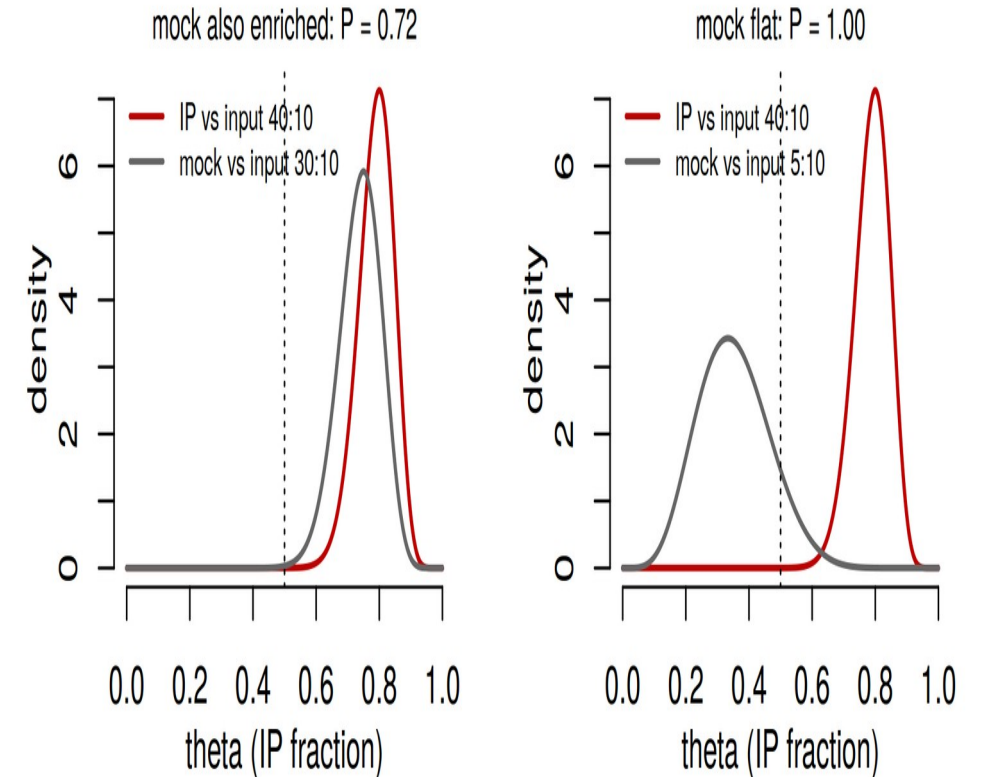
- For a region: n_i reads in IP, n_d in input. The fraction $\theta_1 = IP / (IP + input)$ has a likelihood curve, hence a posterior (Beta). Enriched means $\theta_1 > 0.5$.
- For the mock experiment: n_m reads in mock IP, $n_{d'}$ in its input $\rightarrow \theta_2$. If the mock is also enriched, any antibody pulls this region down: a 'hyper-ChIPable' region.
- **A genuine peak needs $\theta_1 > 0.5$ AND $\theta_1 > \theta_2$.** $P(\text{both} \mid \text{data})$ is the score; rank peaks by it together with the peak caller's score.

Same likelihood curve as 54/138, drawn twice and compared. No new machinery — only the right question.

The posterior by simulation

R

```
set.seed(0)
posterior <- function(n_i, n_d, n_m, n_dm, N = 100000) {
  t1 <- rbeta(N, n_i + 1, n_d + 1)      # IP vs its input
  t2 <- rbeta(N, n_m + 1, n_dm + 1)     # mock vs its input
  mean(t1 > 0.5 & t1 > t2)
}
posterior(40, 10, 30, 10)      # mock also enriched → 0.72
posterior(40, 10, 5, 10)       # mock flat → 1.00
```



Identical IP enrichment (40 vs 10). The mock decides: enriched → doubtful; flat → real. Input alone would call both.

Design the control

YOUR TURN You are doing ChIP-seq for a low-abundance TF in whole fly embryos. Which control, and why?

- A. DNA input only — cheapest, deepest library
- B. Mock IP only — removes non-specific binding
- C. Both: input for calling, mock for scoring

ANSWER C. In fly, 88 % of input-only peaks were spurious; but a mock library alone is too shallow for a low-abundance TF. Input for sensitivity, mock to remove hyper-ChIPable regions.

The toolbox

You have...	Ask...	Concept → tool
k of n reads carry an allele	is $\theta = 0.5$? how sure?	coin → binomial test; likelihood → Beta posterior
reads in a window vs background	more than chance?	rice → Poisson tail test (MACS)
counts × replicates × conditions	which genes change?	donors → NB + borrowed strength + BH (DESeq2)
reads compatible with several isoforms	how much of each?	hidden jars → EM (RSEM)
thousands of p-values	how many are false?	candy colours → FDR, Benjamini–Hochberg
a peak in IP	is it in mock too?	two likelihood curves → joint posterior (Xu 2021)

What you can now do

- Explain a p-value with a coin, and say what it is not.
- Say why 19,271 tests need a correction, and what FDR 5 % promises about your list.
- Say why replicates are not Poisson, and what DESeq2 does about it in four steps.
- Explain the E-step and the M-step with two jars, and run EM in ten lines.
- Explain why a peak needs a control, and what mock IP adds to DNA input.

Tomorrow's lab: the coin test on real mismatches, the 19,271-tests trap, BH by hand.

References

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Questions?

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Course site: Lecture 11 notes with every demo runnable in the browser.